Burn ICU Case Study: Marcus Delaney, a Three-Day ICU Stay, Shift by Shift - FACULTY KEY

Passing score: 78 of 100 points.

How to score (applies to every item)

- Score only what is written in the student's box. Spelling and grammar are not scored. Accept equivalent wording and correct synonyms. Give each element once.
- Partial points: use the amounts printed beside each element. Round the item total to the nearest 0.5. An item never earns more than its maximum.
- If a box is illegible or blank, give 0 and list the item on the score sheet under "faculty review."
- Chart answers are judged against the chart at the time written in the question. Small differences in totals read from the EHR (plus or minus 50 mL) are acceptable.

About this patient

- Marcus Delaney, 34, 82 kg, 180 cm. House fire at 0130 Monday. Flame burn 22% TBSA (right arm circumferential, anterior chest, face and neck; about 8% full thickness) with smoke inhalation (COHb 14.2%), intubated at 0250, right forearm/hand escharotomy at 0315, Burn ICU arrival 0400.
- Parkland: 4 mL x 82 kg x 22% = 7,216 mL lactated Ringer's in 24 hours from the time of injury. Day 2 (Tuesday) he develops burn wound sepsis (MSSA), needs norepinephrine, and improves overnight; he is extubated Wednesday 1045 and transfers Thursday 0700.
- All six shifts are written. The point total is 100: Shift 1 = 20 (it includes the admission picture and the handoff), Shifts 2 to 6 = 16 each.

PART 1: Shift 1: Monday 11/02, 0700 to 1900 (day shift) (20 pts)

Shift 1 is mostly chart reading and the safety reasoning behind burn resuscitation. A careful student should earn about 16 of the 20 points. The hardest items are 1.2 (the Parkland math with the EMS bolus) and 1.10 (the fluid totals).

1.1 Why Is He Here? Your Picture of Marcus at 0700 *[2.5 pts]*

Set the clock to 0700. You are taking report from the night nurse. Read the ED Provider Note, the Burn Surgery and Critical Care H&Ps, the nursing notes from 0425 to 0650 and the Patient Summary. In your own words, explain what happened to Marcus, what injuries he has and what has already been done, then name the THREE most urgent problems you must manage over the next 12 hours.

Required element (accept equivalent wording)	Where / who	Pts
Mechanism and burn: house fire about 0130 Mon, flame burns 22% TBSA (right arm circumferential, anterior chest, face/neck), about 8% full thickness (right forearm and hand).	ED note, Burn H&P	0.5
Inhalation injury and carbon monoxide: hoarse voice, stridor, soot, singed nasal hairs; COHb 14.2% at 0250; intubated 0250; bronchoscopy shows grade 2 inhalation injury.	ED note, bronchoscopy note	0.5
Right forearm/hand escharotomy at 0315 (tense forearm, no Doppler signal) with pulses returned.	Escharotomy note	0.5
Burn shock resuscitation: Parkland lactated Ringer's started at 450 mL/h and titrated to urine output (raised to 540 mL/h at 0600 for urine output of 25 and 26 mL/h); lactate 5.8 at 0250 down to 3.4 at 0415 (the 0800 result comes later).	Orders, labs, nursing note 0605	0.5
Ventilated and sedated (propofol, fentanyl); lines/tubes: left IJ central line, left radial arterial line, OG tube, Foley with temperature probe.	Flowsheets	0.25
Three priorities that make sense, for example: (1) airway/ventilation and carbon monoxide; (2) fluid resuscitation titrated to urine output; (3) circulation to the right hand after escharotomy (also acceptable: pain/sedation, infection prevention).		0.25

1.2 The Parkland Formula *[2 pts]*

Find the burn size and his weight in the chart. (a) Calculate his total 24-hour fluid need with the Parkland formula. (b) How much must he receive in the FIRST 8 hours, and by what clock time does that period end? (c) EMS gave 500 mL of lactated Ringer's before the ED. Explain how that changes the hourly rate for the first 8 hours, counting from his ED arrival at 0230. (d) What rate is the second half of the fluid given at? Show your work with the units. Why is the clock started at the time of the injury and not at his arrival?

Required element (accept equivalent wording)	Where / who	Pts
Total: 4 mL x 82 kg x 22 = 7,216 mL lactated Ringer's in 24 hours (0.25 formula, 0.25 answer).		0.5
First half = 3,608 mL in the first 8 hours from the INJURY (0130 to 0930).		0.5
Subtract the EMS 500 mL: 3,108 mL over the 7 hours from 0230 to 0930 = about 444 mL/h (accept 450 mL/h, which matches the order).	Order: LR 450 mL/h	0.5
Second half = 3,608 mL over the next 16 hours (0930 to 0130 Tue) = 225 mL/h.		0.25
The clock starts at the injury because the capillary leak and fluid loss begin then; the first hours are the most critical (any equivalent reason).		0.25

1.3 Vital Sign Trend *[1.5 pts]*

Copy the vital signs at each time (he has an arterial line, so use the blood pressure shown). Circle every abnormal value. Then, in one or two sentences, describe the trend over the shift and say which value on the flowsheet you do NOT fully trust and why.

Vital	Mon 0700	Mon 1100	Mon 1500	Mon 1900
Temperature	98.2 °F	98.9 °F	99.1 °F	99.2 °F
Heart rate	111	104	102	97
Blood pressure	99/57	105/62	107/62	109/63
Respiratory rate	18	18	17	17
SpO2 (%)	99	99	99	98
Pain (CPOT)	CPOT 1/8	CPOT 1/8	CPOT 1/8	CPOT 1/8

Score 1.0 x (cells correct / 24), rounded to the nearest 0.5. The trend sentence is worth 0.5: heart rate starts high (111) and falls to 97 as the resuscitation works; blood pressure is low-normal (99/57 to 109/63, MAP about 71 to 78); temperature rises from 98.2 F

to 99.2 F (he arrived cold at 96.8 F); CPOT stays 1 because he is sedated. The value NOT to trust fully is the SpO2 of 98-99%: carbon monoxide on the hemoglobin makes the pulse oximeter read falsely normal. Allow +/- 2 for rates and +/- 0.2 F. Abnormal values to circle: heart rate above 100 (111, 104, 102), and temperature, blood pressure and CPOT may be circled or not if the student explains why (accept reasonable choices).

1.3b Trend (scored with 1.3) *[0 pts]*

Describe the trend over the shift and name the value you do NOT fully trust and why.

Model answer: Scored with item 1.3.

1.4 The Pulse Oximeter Says 100% *[1.5 pts]*

Marcus's SpO2 reads 99 to 100% all morning. (a) Explain why this number can be falsely reassuring in this patient. (b) Which lab results show how much carbon monoxide is on his hemoglobin, and what were the values at 0250, 0415 and 0800? (c) What treatment lowers it, and what did the night team change at 0600 (read their notes)? (d) He smokes a pack a day: what does that tell you about the value he is heading toward?

Required element (accept equivalent wording)	Where / who	Pts
A pulse oximeter cannot tell carboxyhemoglobin from oxyhemoglobin, so it reads normal even though the blood carries much less oxygen.		0.5
Carboxyhemoglobin (co-oximetry on the arterial blood gas): 14.2% at 0250, 6.4% at 0415, 3.1% at 0800.	Lab Results	0.25
100% oxygen (FiO2 1.0 through the ventilator) shortens the half-life of carbon monoxide; FiO2 was weaned from 1.0 to 0.60 at 0600 as the level fell (6.4% at 0415), with a repeat level at 0800 before weaning further (FiO2 0.50 at 1400).	Flowsheets, MAR/orders	0.5
A heavy smoker normally runs a COHb of roughly 3 to 10% (accept 3 to 5% or higher), so a level around 3% is at his baseline and not further poisoning; the level is heading down to his smoker baseline, not to zero.		0.25

1.5 Titrating the Fluids *[1.5 pts]*

Open the lactated Ringer's order, the MAR and the hourly urine output. (a) What is the urine output goal in mL/h and in mL/kg/h? (b) At 0930 the rate was changed: from what to what, and why? (c) Using the graph, write the urine outputs for 1200-1300 and 1300-1400, say what the order tells you to do, and what the new rate becomes at 1400. (d) Name two things you check when you change the rate and one danger of giving too much fluid.

Required element (accept equivalent wording)	Where / who	Pts
Goal 30 to 50 mL/h, which is 0.5 mL/kg/h (about 41 mL/h for 82 kg).	Order instructions	0.25
At 0930 (8 hours after the injury) the first half of the Parkland fluid is done, so the rate dropped from 540 to 225 mL/h for the second 16 hours.	MAR	0.25
Urine output 28 mL and 28 mL: below 30 mL/h for two hours in a row, so increase the rate by 20 to 25% and notify the burn team.	I&O, order	0.5
225 mL/h x 1.2 = 270 mL/h at 1400.	MAR	0.25
Two checks (lung sounds / peak airway pressures, abdomen or bladder pressure, right hand pulses and tension, edema, blood pressure, heart rate) AND a danger of too much fluid ("fluid creep": pulmonary edema, compartment syndrome, more ventilator days, worse swelling).		0.25

1.6 Calling the Burn Surgeon *[1.5 pts]*

It is 1400 and you have just seen the second low urine output. Write the SBAR you give Dr. Okafor on the phone. Use real numbers from the chart. End with what you are asking him to do and how you will confirm the order.

Required element (accept equivalent wording)	Where / who	Pts
Situation: identify yourself, the patient and room; urine output 28 mL/h and 28 mL/h for two hours (about 0.34 mL/kg/h, less than the goal of 0.5) on lactated Ringer's 225 mL/h.		0.375
Background: 34-year-old, 82 kg, 22% TBSA flame burn with inhalation injury, intubated and sedated, on the Parkland resuscitation since 0230, EMS 500 mL, escharotomy right arm.		0.375
Assessment: current vitals at 1400 (T 99.0, HR 102, BP 108/63 with MAP 78, SpO2 99%; accept the 1300 values HR 103, BP 105/61), lungs/peak pressures (PIP 27, Pplat 20), bladder pressure 12 mmHg, right hand Doppler pulses, lactate normal at 1.8 (1200), CK 1,120, and your concern that		0.375

he is behind on fluid.		
Recommendation: ask to increase the rate by 20% per the order (to 270 mL/h) and what urine output you should call about next; read the order back and document the time and name.		0.375

1.7 Lines, Tubes and Airway *[1.5 pts]*

List SIX lines, tubes or devices Marcus has at 0700 (not counting the ventilator). For each, write the site and what you check every shift, and one complication you are watching for.

Model answer: Each device row is worth 0.25: the device and site are right AND the check/complication fit that device.

Required element (accept equivalent wording)	Where / who	Pts
Endotracheal tube, 8.0 oral, 24 cm at the teeth: confirm depth and securement, cuff pressure (about 24), HOB 30, oral care, suction as needed. Complications: accidental extubation, VAP, pressure injury to the lips, tube obstruction.		0.25
Left internal jugular central line (triple lumen): chlorhexidine dressing intact, scrub the hub, daily need review, label lumens. Complications: infection (CLABSI), air embolism, clot, bleeding.		0.25
Left radial arterial line (right arm is burned): waveform, level and zero, check the left hand circulation, pressure bag 300 mmHg. Complications: hand ischemia, bleeding or disconnection, infection.		0.25
OG tube 16 Fr: position confirmed by x-ray, mark at the lips, residual and feeds per order, HOB 30. Complications: aspiration, misplacement, intolerance (residual, distention).		0.25
Foley catheter 16 Fr with temperature probe: hourly output, secured, closed system, catheter care. Complications: CAUTI, kinks/low output that is really a blocked catheter, bladder pressure reading for compartment syndrome.		0.25
Left antecubital 18 g peripheral IV: site check, flush, patency. Complications: infiltration, phlebitis, infection. (Also accept the escharotomy sites on the right arm as a device row.)		0.25

1.8 The Ventilator and Sedation *[1.5 pts]*

Open Flowsheets > Respiratory at 0700. (a) Calculate his ideal body weight (male: 50 kg + 2.3 kg for every inch over 5 feet; 1 inch = 2.54 cm) and the tidal volume at 6 mL/kg. Does this match the set tidal volume? (b) List FOUR actions in the ventilator-associated pneumonia (VAP) bundle that you do this shift. (c) What is the RASS goal on Monday, and why is he sedated more deeply on the first day?

Required element (accept equivalent wording)	Where / who	Pts
180 cm = 70.9 inches, so IBW = 50 + 2.3 x 10.9 = about 75 kg; 6 mL/kg = about 450 mL. The set tidal volume is 450 mL, which matches (accept 75-76 kg and 450-455 mL).		0.75
Any four VAP bundle actions: head of bed 30 degrees or higher, oral care with chlorhexidine every 12 h, daily sedation interruption and spontaneous breathing trial, cuff pressure check, suction only when needed, stress ulcer and DVT prophylaxis, hand hygiene (0.125 each).		0.5
RASS goal -2 to -3 on Monday; deeper sedation and analgesia are needed on day 1 for painful burn care, resuscitation, the airway swelling and the risk of pulling the tube.		0.25

1.9 Reading the Labs *[1.5 pts]*

Open Lab Results and compare the first draws (about 0240-0250) with the 1200 and 1800 draws. Choose FOUR labs or lab groups that changed or matter. For each, write the values, why this happens in a burn patient, and what you do about it.

Model answer: Any 4 of the labs below; each row 0.375: values correct (0.125), reason correct (0.125), nursing action reasonable (0.125).

Required element (accept equivalent wording)	Where / who	Pts
Lactate 5.8 -> 3.4 -> 2.2 -> 1.8 -> 1.6 (critical to normal): poor tissue perfusion early; falling means the fluid resuscitation is working. Keep trending; a rise means shock or sepsis, notify.		0.375
Hemoglobin 17.2 -> 13.6 (and hematocrit 51 -> 41): hemoconcentration from plasma leak on arrival, then dilution as fluid is given. Watch for bleeding and new drops.		0.375
CK 480 -> 1,120: muscle and tissue injury; protect the kidneys with adequate urine output and watch for dark urine (myoglobin); report the trend.		0.375
Potassium 4.3 -> 4.5 -> 4.9 -> 5.0: cells release potassium after burn injury; monitor the ECG and		0.375

recheck; do not replace it, and report the rising trend to the provider (no call value is written in the orders, so accept asking the provider for one).		
WBC 12.8 -> 16.1 (1200): stress response to a major burn, NOT an infection yet; use it as a baseline for comparison later.		0.375
Glucose 143 -> 156 -> 172 -> 188 (serum): stress hyperglycemia that keeps rising; check glucose per order and know the call parameters, because an insulin infusion is likely if it stays above 180.		0.375
pH 7.29 -> 7.39 (7.40 at 1600) and bicarbonate 18 -> 24 (arterial gas): metabolic acidosis from poor perfusion corrected with resuscitation.		0.375
Carboxyhemoglobin 14.2 -> 3.0 (also accept if not used in 1.4).		0.375

1.10 Adding Up Your 12 Hours *[1.5 pts]*

Open Flowsheets > Intake/Output and find the Totals section at the bottom (the screenshot shows where). (a) For the DAY SHIFT 0700 to 1900, what are his total intake, total output and net balance? (b) What was his average urine output in mL/h and in mL/kg/h (82 kg)? (c) Using the Parkland rates, how much lactated Ringer's should he have received BY 1900 counting from the injury (the EMS 500 mL and the ED 675 mL before the ICU count), and how much did he actually receive in all (add up the rates and times on the MAR)? (d) What does that tell you?

Required element (accept equivalent wording)	Where / who	Pts
Day shift 0700-1900 as the flowsheet shows it: intake 4,720 mL, output 447 mL, net +4,273 mL (a large positive balance is expected in burn resuscitation). Use the displayed totals; a student who adds the hourly columns by hand may get slightly different numbers (accept if the method is sound).	Flowsheets > Intake/Output > Totals	0.5
Average urine output 447 / 12 = about 37 mL/h = about 0.45 mL/kg/h (accept 37-38 mL/h and 0.45-0.47; just under the 0.5 goal, inside 30-50 mL/h most of the shift).		0.25
Expected by 1900: first 8 h = 3,608 mL plus 9.5 h x 225 = about 2,140 mL, about 5,750 mL. Actual crystalloid from the MAR: EMS 500 + ED 675 + ICU (450 x 2 h + 540 x 3.5 h + 225 x 4.5 h + 270 x 5 h = 5,153) = about 6,330 mL (accept 6,200-6,350; the progress notes quote about 6,190 at 1800 and 6,280 at 1850 with their own rounding).	MAR rates	0.5
He is about 580 mL (10%) over the Parkland estimate: acceptable because the order is to titrate to urine output, but a reminder to watch for fluid creep (edema, airway pressures, bladder pressure) and not to chase a single low urine output.		0.25

1.11 The 1400 Dressing Change *[1.5 pts]*

Read the Burn Wound Assessment note and the 1520 nursing note. (a) Why was hydromorphone given at 1345 and what tool measures pain in an intubated, sedated patient (name its four parts)? (b) Which topical drugs were applied and where? (c) What do you assess at every dressing change? (d) Name two ways you protect him and yourself from infection.

Required element (accept equivalent wording)	Where / who	Pts
Dressing changes are the most painful part of burn care; the opioid is given about 15 minutes BEFORE the procedure so it is working. CPOT has four parts: facial expression, body movements, muscle tension, compliance with the ventilator.		0.5
Silver sulfadiazine 1% cream to the chest and right arm; bacitracin ointment to the face.	MAR	0.25
Wound color, depth, drainage amount and odor, eschar, blisters, edges and any redness beyond the burn margin (mark and time it), skin around the wound, right hand pulses and capillary refill, and pain response.		0.5
Any two: sterile technique, gown/gloves/mask and contact precautions, hand hygiene, dedicated equipment, cleansing the old cream, covering the wound promptly.		0.25

1.12 Handoff at 1900 *[2 pts]*

It is 1900. Write your SBAR handoff for E. Haddad, RN, the night nurse. Use real numbers and name what the night nurse must watch for and do in the first part of the night, including anything coming up in the next few hours.

Required element (accept equivalent wording)	Where / who	Pts
Situation: patient, room, full code, 22% TBSA burn with inhalation injury, intubated and sedated (propofol and fentanyl), current vitals at 1900 (HR 97, BP 109/63, T 99.2, SpO2 98%; accept the 1800 values HR 98, BP 108/61, T 99.1, SpO2 99%), resuscitation going as planned.		0.5

Score only what is written. Flag anything you cannot read for faculty review.

Background: house fire 0130, escharotomy right arm 0315, Parkland (LR 7,216 mL goal in 24 h), lines (central line, arterial line, OG with feeds started at 1000, Foley), COHb down to 3.0% at 1200 (his smoker baseline).		0.5
Assessment: urine output was 28 mL/h around 1200-1400 and corrected after the rate went to 270 mL/h (now 42-46 mL/h); net about +4 L; lactate 1.6; potassium 5.0 and glucose 188 are climbing; right hand Doppler pulses steady; pain controlled; wound care done at 1400.		0.5
Recommendation / to do: hourly urine output with titration per order; hourly right hand checks; the Parkland 24 hours ends at 0130 Tuesday and the order changes then; 2000 ABG and 0000 labs and CK (glucose may need an insulin infusion if still over 180; recheck potassium); sedation goal and restraint checks; turning and skin care; family visiting; call the burn team for UOP under 20 mL/h.		0.5

PART 2: Shift 2: Monday 11/02 1900 to Tuesday 11/03 0700 (night shift) (16 pts)

Shift 2 tests the "quiet" overnight judgment: the end of the Parkland period (1.2 math repeated with the final total), the insulin infusion, an agitation event in a ventilated patient with an alcohol history, and compartment checks. A careful student should earn about 12 of the 16 points.

2.1 Vital Sign Trend Through the Night *[1.5 pts]*

Set the clock to 1900 on Monday. You take report from the day nurse. Copy the vital signs at each time, circle every abnormal value, then write two sentences below: what the night looked like overall, and what the flowsheet does NOT show about the agitation event that happened around 0245.

Vital	Mon 1900	Mon 2300	Tue 0300	Tue 0700
Temperature	99.2 °F	99.3 °F	99.3 °F	99.4 °F
Heart rate	97	97	93	96
Blood pressure	109/63	109/62	109/64	111/64
Respiratory rate	17	16	16	17
SpO2 (%)	98	99	98	98
Pain (CPOT)	CPOT 1/8	CPOT 1/8	CPOT 1/8	CPOT 1/8

Score 1.0 x (cells correct / 24), rounded to the nearest 0.5. The two sentences are worth 0.5: (1) a quiet night: heart rate 93-97 (upper normal, never over 100), BP about 109-111/62-64 (MAP 78-80), temperature 99.2-99.4 F, SpO2 98-99%, CPOT 1; (2) the hourly flowsheet does not show the 0245 agitation: the vital signs at 0300 (HR 93, BP 109/64, SpO2 98%, CPOT 1) look normal and RASS is charted only every 4 hours (-3 at 0000, -2 at 0400), so an event between hourly sets (RASS +1 at 0245, back to -3 by 0315, lorazepam 1 mg at 0300) is recorded only in the nursing note and the MAR. Allow +/- 2 for rates and +/- 0.2 F.

2.1b Overall night, and what the vitals do not show (scored with 2.1) *[0 pts]*

Model answer: Scored with item 2.1.

2.2 Glucose and the Insulin Infusion *[2 pts]*

Open Lab Results (Glucose, point of care and serum), Orders, the MAR and the 0150-0320 nursing notes. (a) List his glucose values from 1800 through 0300 and explain why the insulin infusion was started at 0030 and at what rate. (b) Why does a burn patient have a high glucose even though he has no diabetes? (c) What is the glucose goal and how often do you check it on the infusion? (d) Give TWO safety actions for an insulin infusion, including what you must do if the tube feeding is stopped, and name one lab that you watch closely with insulin.

Required element (accept equivalent wording)	Where / who	Pts
Glucose: 188 (1800), 198 (2200), 196 (0000), then 191, 190, 189 on the infusion. Started at 1 unit/h at 0030 because the glucose was above 180 on consecutive checks (ICU protocol), ordered by Dr. Brandt.	Labs, notes	0.5
Stress response of a major burn: stress hormones (cortisol, catecholamines, glucagon) and inflammation make the body insulin resistant and the liver release glucose; the hypermetabolic state and the tube feeding add to it.		0.5
Goal 140 to 180 mg/dL; glucose every hour while on the infusion, and adjust the rate per the protocol table.	Order	0.5
Any two safety actions: independent double check of the insulin bag and rate (high-alert drug); do not change the rate without the protocol; if feeds are stopped or interrupted, call the provider and recheck the glucose because hypoglycemia is likely (treat glucose under 70 per protocol); dedicated line/label; check glucose after any rate change. AND one lab watched: potassium (insulin moves potassium into cells; K 5.0 at 1800, 4.5 at 0000, 3.9 at 0600).		0.5

2.3 The End of the First 24 Hours *[1.5 pts]*

The Parkland period ends at 0130 (24 hours after the injury). Using the lactated Ringer's rates and times on the MAR (EMS 500 mL, ED 450 mL/h from 0230 to 0400, then the ICU rates), (a) add up the total crystalloid he received from the injury to 0130 and compare it with the Parkland goal. (b) What did the order change to at 0130 and what new fluid starts at 0600? (c) Why do the fluids not simply stop at 24 hours? (d) His weight was 82 kg on arrival and is now 89.6 kg: what does that tell you and what else do you watch for?

Score only what is written. Flag anything you cannot read for faculty review.

Required element (accept equivalent wording)	Where / who	Pts
Total: 500 (EMS) + 675 (ED 0230-0400) + 900 (0400-0600 at 450) + 1,890 (0600-0930 at 540) + 1,012 (0930-1400 at 225) + 3,105 (1400-0130 at 270) = about 8,080 mL, about 864 mL (12%) over the 7,216 mL goal (accept 8,000-8,150).	MAR	0.5
At 0130 lactated Ringer's was lowered to a maintenance rate of 125 mL/h; albumin 5% 500 mL over 4 hours starts at 0600 per the burn protocol.	Orders	0.375
The capillary leak and losses continue for 24 to 48 hours, so he still needs fluid; colloid (albumin) holds fluid in the vessels and lets you use less crystalloid.		0.25
Weight is up 7.6 kg, so the extra fluid is in the tissues (edema); also watch lung sounds and ventilator pressures, bladder pressure, the right arm compartments and urine output.		0.375

2.4 Alcohol and the Next Three Days *[1 pts]*

His ethanol level was 42 mg/dL at 0240 and he usually drinks 4 to 6 beers on weekends, more than that the night of the fire. (a) When could alcohol withdrawal begin, and when are seizures and delirium tremens most likely? (b) Why may you not see the signs yet? (c) Which two vitamins are ordered and why? (d) Name three signs you would watch for.

Required element (accept equivalent wording)	Where / who	Pts
Withdrawal can begin 6 to 24 hours after the last drink; seizures 12 to 48 hours; delirium tremens 48 to 96 hours, so counting from the last drink (about the time of the fire, Monday 0130) the highest risk for him is Wednesday to Friday.		0.25
Propofol, fentanyl and lorazepam are sedating and mask the signs; and he cannot talk.		0.25
Thiamine (prevents Wernicke encephalopathy, needed before or with glucose) and folic acid (deficiency in alcohol use).	MAR	0.25
Any three: tachycardia, hypertension, fever, tremor, sweating, agitation, hallucinations, seizure, difficulty sedating (needing more drug); use RASS and CAM-ICU because CIWA needs a patient who can talk.		0.25

2.5 The Agitation Event at 0245 *[2 pts]*

Read the 0320 nursing note. Marcus (RASS -3 earlier) becomes restless, thrashes his head, bites the endotracheal tube and the ventilator pressure alarms sound. (a) In order, what do you check and do FIRST, before any medication, and why? (b) The provider orders lorazepam 1 mg IV. Name one concern about giving a benzodiazepine to this patient and one question you could ask the provider. (c) What are the rules for the soft wrist restraint you are using? (d) Besides rechecking RASS, CPOT and vital signs after the dose, what do you document and who do you notify?

Required element (accept equivalent wording)	Where / who	Pts
Stay with him and check for causes before sedating: airway/tube (depth 24 cm, cuff pressure, secretions, kinks, bite block), ventilator alarms and synchrony (call RT), oxygenation, pain (CPOT 1-2), full bladder or Foley problem, positioning, and withdrawal/delirium; keep him safe (no unplanned extubation). Give analgesia first if pain is possible.	Nursing note	0.75
Concern: benzodiazepines increase delirium, prolong ventilation and can mask withdrawal; the unit prefers analgesia-first and propofol/dexmedetomidine. A good question: can we increase the propofol or fentanyl instead, or is a one-time lorazepam dose acceptable while he is ventilated (reason: alcohol-withdrawal risk)? Lorazepam is high-alert: double check the dose.		0.5
Restraint rules (per the order and hospital policy): provider order that is renewed every calendar day, least restrictive method, check circulation and skin at least every 2 hours (the order says q2h; many policies also require release and range of motion), assess the continuing need, document, remove at extubation, family informed.	Order	0.5
Document the event (time, behavior, cause search, response, RASS/CPOT before and after, lorazepam 1 mg IV at 0300 with the reason), notify Dr. Brandt (on-call intensivist, SBAR; done at 0255) and the charge nurse (B. Sutton), and update Jenna at the morning call.		0.25

2.6 Checking the Right Hand and the Abdomen *[1.5 pts]*

Compartment syndrome is a risk after a circumferential burn and a large fluid resuscitation. For each of the six Ps, write what you can or cannot check in Marcus (he is sedated and intubated) and what the chart showed at 2300-0300. Then name the abdominal check ordered every six hours.

Score only what is written. Flag anything you cannot read for faculty review.

Model answer: Rows 1-6 worth 0.2 each (the P is named, a sensible way to check, and the chart finding); row 7 (bladder / intra-abdominal pressure) 0.3.

Required element (accept equivalent wording)	Where / who	Pts
Pain: out of proportion, with passive stretch; cannot be reported while sedated, so look for grimace/CPOT rise and a tense forearm. Chart: CPOT 1 (forearm soft).		0.2
Pallor (and cool temperature): fingers pink and warm. Chart: warm, cap refill 3 seconds.		0.2
Pulselessness: Doppler signals over the radial and ulnar arteries (cannot rely on palpation through edema). Chart: biphasic Doppler signals present.		0.2
Paresthesia: cannot be asked; sensation is not assessable while sedated. Chart: left hand not testable (sedated); right hand decreased in the burned area.		0.2
Paralysis: motion of the fingers, only when sedation is lightened (SAT at 0800). Chart: not testable yet.		0.2
Poikilothermia / tension: skin temperature compared with the left hand, and firmness of the forearm compartments. Chart: fingers warm; forearm compartments soft and compressible, swelling stable.		0.2
Bladder (intra-abdominal) pressure every 6 hours through the Foley: 12 mmHg at 2330; call the provider above 15 (abdominal compartment syndrome is above 20 with organ dysfunction).	Order, note	0.3

2.7 Tube Feeding and Calories *[1.5 pts]*

Open the tube feeding order and the flowsheet. (a) Write the advancing schedule from 1000 Monday and the rate at 0600 on Tuesday. (b) The formula is Osmolite 1.5, which has 1.5 kcal per mL and 62.7 g of protein per liter: how many calories and grams of protein will he receive in 24 hours at 70 mL/h? (c) Give two reasons burn patients are fed early through the gut. (d) What do you check before advancing, and what do you do for a residual over 500 mL?

Required element (accept equivalent wording)	Where / who	Pts
Start 20 mL/h at 1000, then increase 10 mL/h every 4 hours (30 at 1400, 40 at 1800, 50 at 2200, 60 at 0200) to the goal of 70 mL/h at 0600.	Order	0.25
70 mL/h x 24 h = 1,680 mL; x 1.5 kcal/mL = 2,520 kcal; x 62.7 g/L = about 105 g protein (accept 100-110).		0.5
Any two: burns make the body hypermetabolic (needs about 2,400-2,500 kcal and extra protein); early feeding protects the gut lining and lowers bacterial translocation and infection; supports wound healing; reduces muscle loss and stress ulcers.		0.5
Check the position of the tube (OG at the mark), the residual, abdominal distention, bowel sounds, stool, glucose, and keep the head of the bed at 30 degrees; for a residual over 500 mL hold the feeding and call the provider (and do not hold feeds for dressing changes).		0.25

2.8 The Midnight and 0600 Labs *[1.5 pts]*

Open Lab Results and compare the 0000 and 0600 draws (and the 0500 albumin and coagulation labs) with the day shift values. Choose FOUR results. Write the values, why they happen in a burn patient on propofol and tube feeding, and what you do or ask for.

Model answer: Any 4 rows of the bank; each row 0.375: values correct (0.125), reason (0.125), action (0.125).

Required element (accept equivalent wording)	Where / who	Pts
Phosphorus 3.0 (1200) -> 2.8 (1800) -> 2.4 (0000) -> 2.1 (0600) (low): burn losses, insulin and feeding drive phosphate into cells (refeeding), dilution. Report to the provider; replacement is usually ordered; low phosphorus weakens breathing muscles and the heart.		0.375
Magnesium 1.8 (1800) -> 1.7 -> 1.6 (low): dilution and losses; report for replacement (low magnesium makes low potassium hard to correct, so replace magnesium along with potassium).		0.375
Potassium 4.5 -> 3.9: the insulin infusion and feeds move potassium into cells; many protocols aim for 4.0 or higher; notify the provider and recheck with the next draw (K 5.0 at 1800).		0.375
Albumin 2.6 (low) and prealbumin 7.1 (low): plasma leak, dilution and the acute-phase response; they do not measure nutrition well in the first days of a burn, but explain the albumin 5% order and the dietitian follow-up.		0.375
Triglycerides 168 (high; baseline at about 48 h of propofol): propofol is a lipid emulsion (1.1		0.375

kcal/mL); report it, count the calories, and a level over about 400 means ask whether to change the sedative.		
CK 1,120 -> 840 (falling): muscle injury is improving; keep adequate urine output.		0.375
Hemoglobin 12.4 and platelets 214: dilution; platelets still normal; watch for bleeding with heparin prophylaxis. WBC 15.2 is stress response, a baseline for later comparison.		0.375
Sodium 134 (low): lactated Ringer's has less sodium than the plasma and he is retaining fluid; watch; no correction needed now.		0.375

2.9 Propofol: Math and Safety *[1.5 pts]*

At 1900 on Monday the propofol is running at 35 mcg/kg/min. The bag is propofol 10 mg/mL and his weight is 82 kg. (a) Calculate the rate in mL/h. (b) Propofol gives 1.1 kcal per mL: how many calories does the sedation add in 24 hours? (c) Name two reasons you check the triglyceride level and two signs of propofol infusion syndrome. (d) Name two nursing actions for the bag and tubing.

Required element (accept equivalent wording)	Where / who	Pts
35 mcg/kg/min x 82 kg = 2,870 mcg/min x 60 = 172,200 mcg/h = 172.2 mg/h; at 10 mg/mL = 17.2 mL/h.		0.5
17.2 mL/h x 24 h = about 413 mL/day x 1.1 kcal = about 455 kcal/day (accept 450-460); count it as part of his calories.		0.25
Triglycerides: lipid load can raise them (his 168 is mildly high) and very high levels cause pancreatitis; check about every 2 to 3 days. Infusion syndrome: unexplained metabolic acidosis, rhabdomyolysis/rising CK, hyperkalemia, arrhythmia or heart failure, high triglycerides, kidney injury (any two).		0.5
Any two: aseptic technique and change the tubing and vial/bag every 12 hours (lipid grows bacteria), label, watch blood pressure with every rate increase, do not mix other drugs in the line, daily sedation interruption.		0.25

2.10 Handoff at 0650 *[2 pts]*

It is 0650 on Tuesday. Write your SBAR handoff for J. Morales, RN, the day nurse. Use real numbers, include the events of the night, what is happening in the next few hours (look at the orders and MAR), and anything you would like the day team to follow up.

Required element (accept equivalent wording)	Where / who	Pts
Situation: patient/room, full code, burn 22% with inhalation injury, intubated; stable night: HR 96, BP 111/64, T 99.4, SpO2 98%, RASS -2 (goal -1 to -2 today), CPOT 1; urine output 37-47 mL/h (0.45-0.57 mL/kg/h).		0.5
Background: the first 24 hours of Parkland finished at 0130 (about 8,080 mL), LR 125 mL/h from 0130, then 75 mL/h since 0600 while albumin 5% runs; insulin infusion since 0030; feeds at goal 70 mL/h; weight 89.6 kg; the flowsheet shows the night shift (1900-0700) intake 3,748 mL, output 496 mL, net +3,252 mL.	Flowsheets > Intake/Output > Totals	0.5
Assessment: agitation at 0245 treated with lorazepam 1 mg (calm since); glucose 179-191 on the infusion (1.5 units/h since 0600); phosphorus 2.1 and magnesium 1.6 low and potassium down from 5.0 to 3.9, reported to Dr. Brandt, replacement planned after the noon labs and not given overnight; right hand signals steady; bladder pressure 12; triglycerides 168; skin and turns done.		0.5
Recommendation / to do: sedation interruption at 0800; noon labs and replace phosphorus, magnesium and potassium as ordered; hourly glucose; albumin 5% 500 mL running 0600-1000; dressing change and silver sulfadiazine at 1400 with hydromorphone beforehand; restraint order renewal; update Jenna and Linda; chest x-ray (0515: small new right lower lobe atelectasis, lines in place) so chest PT and turning; watch for alcohol withdrawal.		0.5

PART 3: Shift 3: Tuesday 11/03 0700 to 1900 (day shift) (16 pts)

Shift 3 follows the safe wake-up, the electrolyte orders, early recognition of burn wound infection (graph analysis, SBAR), the sepsis bundle with antibiotic checks, the norepinephrine math and the shock plan. A careful student should earn about 12 of the 16 points.

3.1 The 0800 Sedation Interruption *[1 pts]*

Set the clock to 0700 and read the 0850 nursing note and the 0845 respiratory therapy note. (a) What was done at 0800 and what was the order for the sedatives? (b) How did Marcus respond (RASS, commands, CAM-ICU, his mother)? (c) Name two findings that would make you stop the interruption early and restart the sedation. (d) Why is he NOT having a breathing trial (SBT) today even though FiO2 is 0.40 and PEEP is 5?

Required element (accept equivalent wording)	Where / who	Pts
Propofol held at 0800 (was 30 mcg/kg/min) while fentanyl 100 mcg/h continued for pain; the order says wake to RASS 0 to -1 or until agitation or distress, then restart at half the prior rate (restarted 0835 at 20 mcg/kg/min, back to 30 at 0930).	Orders, MAR, note	0.25
Woke to voice within minutes, RASS 0, followed commands with the LEFT hand and left leg, mouthed "mom", tearful, calmed holding his mother Linda's hand; CAM-ICU negative; HR 92-98, MAP 78-82, SpO2 98%, no dyssynchrony.	Nursing note 0850	0.25
Any two stop criteria: agitation or distress, pulling at the tube or lines, ventilator dyssynchrony or biting, SpO2 under about 88-90%, sustained HR/BP change, respiratory rate over about 35, seizure or new arrhythmia, unsafe pain.		0.25
Still in the post-resuscitation fluid-shift phase (albumin 5% running until 1000, Parkland ended only at 0130), airway/facial edema, no cuff-leak test, a 1400 dressing change needing sedation and analgesia; reassess tomorrow (Dr. Brandt and RT).	Progress note, RT note	0.25

3.2 Albumin and the Feeds *[1 pts]*

Albumin 5% 500 mL runs from 0600 to 1000. (a) Calculate the pump rate in mL/h. (b) Why is albumin given to Marcus now (use a lab value), and what two things do you assess during the infusion? (c) What did the LR rate change to at 1000? (d) The dietitian wrote about the tube feeding and the 1400 dressing change: what did she recommend about holding the feeding?

Required element (accept equivalent wording)	Where / who	Pts
500 mL / 4 h = 125 mL/h.		0.25
Serum albumin 2.6 (low) and ongoing capillary leak: albumin holds fluid in the vessels, so less crystalloid is needed (he is already about 12% over the Parkland goal). Assess for fluid overload (lung sounds, SpO2, ventilator pressures, edema) and for reaction (fever, chills, hypotension, rash) plus urine output.	Labs, notes	0.5
LR went from 75 mL/h to 100 mL/h at 1000 (titrate urine output 0.5-1 mL/kg/h).	MAR	0.125
Do NOT hold the feeding (Osmolite 1.5 at goal 70 mL/h) for dressing changes; hold only for the OR or extubation (and she added a protein modular 15 g twice a day).	RD note	0.125

3.3 Noon Electrolytes and the Replacement Orders *[1.5 pts]*

Open the 1200 labs, the orders and the MAR. For each result, write the value, the replacement ordered (drug, dose, route, rate) with the time you gave it, and the safety check or monitoring you do. Then answer below the table: why did the potassium and phosphorus drop when he is on an insulin infusion and tube feeding, and what other 1200 result could the phosphate make worse?

Model answer: Rows 0.4 each (value 0.1, order and time 0.15, safety 0.15) = 1.2; the two sentences below the table 0.3.

Required element (accept equivalent wording)	Where / who	Pts
Potassium 3.6 (goal 4.0 or higher): KCl 20 mEq in 100 mL over 2 h (10 mEq/h) through the central line, given 1230 and a second dose 1430. Safety: central line only, maximum 10 mEq/h, telemetry, separate pump, double check; hold and call for urine output under 30 mL/h for 2 hours; recheck K 4 h later.	Orders, MAR	0.4
Magnesium 1.7: magnesium sulfate 2 g in 50 mL over 1 h IV, given 1330. Safety: watch BP, HR and reflexes/respiration (hypotension, flushing); recheck level with the evening labs.	MAR	0.4
Phosphorus 2.1 (low): sodium phosphate 15 mmol in 250 mL over 4 h via the central line, started 1500. Safety: slow infusion, separate pump, watch calcium (7.6, low) and renal function; recheck	MAR	0.4

Score only what is written. Flag anything you cannot read for faculty review.

with the evening labs.		
Insulin and feeding (refeeding) drive K and phosphorus into the cells, plus burn and fluid losses and dilution. Phosphate binds calcium and can lower an already low calcium (Ca 7.6): watch for tingling, cramps, QT change and ask about the calcium.		0.3

3.4 The 1400 Dressing Change *[1 pts]*

You and N. Ibarra, RN (burn nurse specialist) change the dressings. Read the 1425 and 1455 notes. (a) List the four NEW findings that told you this is not a normal burn wound. (b) What pain medicine was given, when, and how did CPOT change? (c) What did you do to the edge of the red area at 1430 and why? (d) Which finding at the right chest border is the most worrisome for tissue death?

Required element (accept equivalent wording)	Where / who	Pts
New since Monday: purulent drainage (thick yellow-green) on the chest/arm dressings; foul odor; erythema spreading beyond the burn margin (about 2-3 cm in the notes, 4 cm on the 1430 flowsheet entry; warm, indurated); eschar separating with a dark violaceous area. (Any four of these, including increased pain with cleansing.)	Notes	0.5
Hydromorphone 0.5 mg IV at 1345, 15 minutes before the change; CPOT 2/8 at 1300, 5/8 during dressing removal, back to 4 by 1500.	MAR	0.25
Marked the margin of erythema with a skin marker (dated and timed 1430), measured and photographed, to see whether it spreads; recheck every 4 hours and every shift.		0.125
The dark violaceous, non-blanching area about 4 x 5 cm at the right chest border with loss of eschar adherence: possible invasive infection with tissue necrosis (needs surgeon exam and biopsy).		0.125

3.5 Analyze the Graph: Temperature, Heart Rate and MAP *[1.5 pts]*

Study the figure (it is drawn from the flowsheet for 0700 to 1900). (a) At about what time did the change begin and which of the three values moved first? (b) Describe the pattern at 1400, 1700 and 1900 in your own words and say what it means for Marcus. (c) After the norepinephrine starts at 1730 the MAP climbs back up: what does the graph tell you about whether he is getting better? (d) Burn patients often have a fast heart rate and a temperature near 100 F even when they are well: why are the usual SIRS criteria unreliable for him, and name two signs you use instead.

Required element (accept equivalent wording)	Where / who	Pts
Stable all morning (T 99.4-99.7, HR 92-98, MAP 77-80); the change starts about 1300 (T 99.9, a slow creep from 1100), and the sharp bend is between 1300 and 1400. Temperature moved first (slowly, from 99.7 to 99.9); then HR (97 to 105) and MAP (77 to 72) changed together at 1400.	Graph, vitals	0.5
Steady rise of temperature (100.4 at 1400, 101.6 at 1700, 102.3 at 1900) and HR (105, 122, 130) with falling MAP (72, 66, then 63 at 1715, below the goal of 65): the picture of evolving sepsis from the wound infection, going into septic shock (vasodilation and low perfusion); it is a trend, not a single value.	Graph	0.5
MAP recovers (64 to 70) because the drug squeezes the vessels, but temperature and heart rate keep climbing (HR 130, T 102.3), so the infection is not controlled; he is not better, the pressor treats the number, not the cause (needs source control: antibiotics, surgery, fluids carefully).	Graph	0.25
He is already hypermetabolic after the burn (baseline HR above 90 and temperature 99-100 F, tachypnea, high WBC and glucose), so SIRS criteria are met by almost every burn patient and do not separate sepsis. Better: trend (progressive tachycardia over 110, temperature over about 102 F/39 C, RR over 25), new hyperglycemia/insulin resistance (glucose 205-244), falling platelets, feeding intolerance, falling urine output, rising lactate, wound changes. (Accept any two.)	Event note 1755	0.25

3.6 The SBAR Call at 1540 *[1.5 pts]*

It is 1540. You are going to call Dr. Okafor and Dr. Brandt. Write what you say (S, B, A, R) using the chart numbers at 1530 and the trend since 1300. End with what you ask them for: name at least three orders or decisions you request.

Required element (accept equivalent wording)	Where / who	Pts
Situation: bed/name, suspected burn wound infection with sepsis; 1530 T 101.1 F, HR 113, BP 105/51 (MAP 69), RR 23, SpO2 96% on the ventilator.	Vitals	0.375
Background: 34-year-old man, 22% TBSA flame burn with inhalation injury, post-burn day 2,	Notes	0.375

Score only what is written. Flag anything you cannot read for faculty review.

intubated, NKDA, full code; 1400 dressing change findings (purulent drainage, foul odor, spreading erythema, violaceous eschar at the right chest border).		
Assessment: trend T 99.9 (1300), 100.4, 100.9, 101.1; HR 97, 105, 108, 113; MAP 77, 72, 71, 69; RR 18 to 23; urine 45, 41, 36 mL/h (0.44 mL/kg/h, falling); glucose 205 at 1530 (was 173); WBC 15.4 and lactate 1.4 earlier; CPOT 4. "I am concerned this is early sepsis."	Flowsheets	0.375
Recommendation: bedside wound exam by the surgeon; sepsis order set: cultures before antibiotics (blood x2, wound, tracheal aspirate, urine), lactate, procalcitonin, CBC with differential, coags; broad-spectrum antibiotics after cultures; fluids or MAP plan; call parameters (MAP under 65, urine under 30 mL/h). (Any three requests.)	Note 1550	0.375

3.7 The Sepsis Bundle: Cultures and Lactate *[1.5 pts]*

Open the 1655 nursing note and Lab Results. (a) List the four culture sites (five specimens) and the time collected, and which two blood sites. (b) Why must the cultures be drawn BEFORE the antibiotics, and when is it OK to give antibiotics without waiting? (c) Write the lactate values and times from 1555 to 1700 and what they mean. (d) What is the goal for giving antibiotics after recognizing sepsis, and what were Marcus's actual times?

Required element (accept equivalent wording)	Where / who	Pts
Collected 1550: blood culture set 1 peripheral (left hand), set 2 from the left IJ central line, wound swab (right chest/arm, after cleansing, from the drainage), tracheal aspirate, urine from the Foley port; all labeled and sent. (Wound biopsy 1635 by Dr. Okafor.)	Note 1655	0.5
Antibiotics kill or suppress the organisms and a culture drawn afterward can be falsely negative, so you could not target or narrow therapy; draw before the first dose, but in septic shock the antibiotics must not be delayed (goal within 1 hour); if cultures cannot be obtained quickly, give the antibiotics and document why.		0.5
Lactate 2.9 at 1555 (high), 3.8 at 1700 (rising): tissue hypoperfusion (anaerobic metabolism); the rise despite maintenance fluids (the first bolus did not start until 1700) means he is getting worse, repeat in 2 hours (goal clearance 10-20% every 2 hours). Also WBC 18.4, procalcitonin 4.8 at 1555.	Labs	0.25
Antibiotics within 1 hour of recognition: recognized and SBAR 1540, cultures 1550, pip-tazo 1630 (about 50 minutes after the call), vancomycin 1645 (about 65 minutes).	MAR	0.25

3.8 Checking the Antibiotic Orders *[1.5 pts]*

Before you hang the antibiotics you check the orders with pharmacy (A. Chen, PharmD). Marcus weighs 82 kg. (a) The vancomycin load is 20 mg/kg: calculate the dose and compare it with the 1,750 mg on the MAR, then give the pump rate in mg/h for the 2-hour infusion and the order's maximum rate. (b) The piperacillin-tazobactam 4.5 g in 100 mL is an extended infusion: over how long, at what mL/h, and why is it given this way? (c) What does the pip-tazo order say about running it with the vancomycin, and what will you do with the lumens of the left IJ? (d) Name two lab values you follow with this combination and why.

Required element (accept equivalent wording)	Where / who	Pts
20 mg/kg x 82 kg = 1,640 mg; the 1,750 mg is 21.3 mg/kg, about 7% above 1,640 mg (the order says "about 20 mg/kg, rounded to 1,750 mg", the nearest 250 mg; pharmacy verified). 1,750 mg / 2 h = 875 mg/h, under the order maximum of 1 g/h (slow the rate if red man syndrome).	Orders, note	0.5
Infuse over 4 hours: 100 mL / 4 h = 25 mL/h. Extended infusion keeps the drug level above the organism MIC for longer (time-dependent beta-lactam), which helps in sepsis and with Pseudomonas.	Order	0.375
The order says to give pip-tazo through its own lumen when possible, separate from vancomycin, and that the combination is associated with kidney injury (pharmacy also advises no co-infusion because compatibility depends on the concentration); run them on separate lumens/pumps of the left IJ (the norepinephrine later gets its own lumen too, never shared), and ask pharmacy if lumens run short.	Order, note	0.375
Creatinine (1.1 at 1700; vancomycin and pip-tazo are nephrotoxic, and his urine output is falling) and vancomycin levels (pharmacy AUC dosing); also watch WBC/temperature for response and check the cultures to narrow in 48 hours.	Labs	0.25

3.9 Low MAP and Norepinephrine *[1.5 pts]*

At 1715 the BP is 101/44 (MAP 63-64) after the 1700 LR bolus; the second bolus is given at 1745 and norepinephrine 16 mcg/mL is started at 1730 at 0.05 mcg/kg/min. (a) Calculate the pump rate in mL/h for 0.05 mcg/kg/min (82 kg), and the rate at 0.10 mcg/kg/min that you reached at 1830. (b) Where must it run and why (name the complication and one sign)? (c) How and how fast do you titrate, and what is the MAP goal and how is the MAP measured? (d) Why do the providers want to limit more fluid boluses?

Required element (accept equivalent wording)	Where / who	Pts
0.05 x 82 = 4.1 mcg/min x 60 = 246 mcg/h / 16 mcg/mL = 15.4 mL/h. At 0.10: 8.2 mcg/min = 492 mcg/h = 30.8 mL/h (as printed on the order).	Order	0.5
Central line only, on a dedicated lumen with nothing else in it (never a PIV): extravasation causes tissue necrosis; check the site and the hand/extremity every hour (pain, blanching, swelling, cool skin, loss of blood return); if suspected stop the drip and notify the provider at once (the provider may order phentolamine; follow the unit extravasation protocol).	Order	0.375
Titrate by 0.02 mcg/kg/min every 5 minutes (order; max 0.5 without attending approval) to MAP 65 or higher; wean by 0.02 every 15-30 minutes when MAP is over 75; MAP read from the left radial arterial line (leveled and zeroed); give a second RN check for each change; she went 0.05, 0.08 (1800), 0.10 (1830).	Order, MAR	0.375
He is already about 12% over the Parkland total with moderate edema, the rising airway pressures (PIP 24 to 28, Pplat 17 to 21) and SpO2 falling from 98% to 95% suggest leak into the lungs, and the boluses lifted MAP only briefly; more fluid worsens edema and ARDS risk. Give vasopressor instead.	Event note	0.25

3.10 Switching the Topical Antimicrobial *[1 pts]*

Silver sulfadiazine was stopped and mafenide acetate 5% soaks were started at 1730. (a) Why did the burn surgeon choose mafenide for this wound? (b) Name the two things you monitor with mafenide and look at the 1700 bicarbonate and the 1845 arterial blood gas: what do they show and why can you not blame the mafenide alone? (c) What do you give before each soak and how often are they done?

Required element (accept equivalent wording)	Where / who	Pts
Mafenide penetrates eschar and wound tissue (the infection is under separating eschar), unlike silver sulfadiazine which does not; it is active against the Gram-negatives and many organisms in the wound.	Event note	0.25
Monitor metabolic acidosis (mafenide is a carbonic anhydrase inhibitor: BMP/bicarbonate and ABG) and pain/burning on application. Chart: CO2 23 (1200) to 22 (1700); ABG 1845 pH 7.31, HCO3 18, base excess -8, PaCO2 36. This is a real metabolic acidosis, but lactate was already 3.8 at 1700 (septic shock) so the cause is mainly lactic acidosis, and mafenide can add to it: report it and trend.	Labs	0.5
Pain medicine (hydromorphone/fentanyl) before the soak; soaks every 8 hours (next 0130), keep the gauze moist.	Order	0.25

3.11 Falling Urine Output *[1 pts]*

The urine output was 45, 41, 36, 28, 24 and 20 mL per hour from 1200 to 1800 (his goal is 0.5 mL/kg/h = 41 mL/h, and Foley care is done). At 1800 the output is 20 mL (0.24 mL/kg/h). Which actions are correct for you now?

Correct: Report it to Dr. Brandt and Dr. Okafor now...; Check the Foley for kinks...; Hold any further potassium replacement and tell the provider...; Ask about the doses of nephrotoxic drugs...; Keep recording hourly urine output and total I&O...

About 0.2 per correct option selected (5 correct = 1.0); subtract 0.2 for each wrong option selected (not below 0). Wrong: asking for another bolus without assessing (he is 12% over Parkland, edema, new lung findings and rising pressures; the oliguria is from low perfusion/sepsis), stopping the norepinephrine (MAP 63-64 would fall), a diuretic in septic shock (would worsen perfusion and kidney injury). By 1900 the chart totals for the shift are 4,094 in, 474 out, +3,620 net.

Common errors: Treating the number (more fluid or a diuretic) without looking for the cause, or not reporting it.

3.12 Handoff at 1850 *[2 pts]*

It is 1850. Write your SBAR handoff for E. Haddad, RN, the night nurse. Use real numbers from your shift, say what is changing, and list what the night nurse must do or follow up in the next hours (look at the orders, MAR and labs).

Score only what is written. Flag anything you cannot read for faculty review.

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Required element (accept equivalent wording)	Where / who	Pts
Situation: septic shock from a right chest/arm burn wound infection; 1830 T 102.1 F, HR 126, BP 108/48 (MAP 68), RR 29, SpO2 94%; on norepinephrine 0.10 mcg/kg/min (30.8 mL/h) to keep MAP 65 or higher; full code, NKDA.	Vitals, note	0.5
Background: 22% TBSA flame burn with inhalation injury, post-burn day 2, intubated; Parkland ended 0130; wound infection found at 1400; cultures x4 sites 1550; pip-tazo 1630 (extended, 4 h) and vancomycin 1,750 mg 1645 (until about 1845) on separate lumens; LR 500 mL x2 (1700, 1745); norepinephrine started 1730 (0.05, 0.08 at 1800, 0.10 at 1830); mafenide started 1730, silver sulfadiazine stopped; biopsy 1635.	MAR	0.5
Assessment: lactate 2.9 then 3.8 (repeat due 1900); ABG 1845 pH 7.31, HCO3 18, BE -8; urine 20 mL/h (0.24 mL/kg/h); glucose 244 on insulin 4 units/h (feeds still 70 mL/h); ventilator changed at 1800 to PEEP 8, FiO2 0.50, rate 18, PIP 29/Pplat 22 (early ARDS physiology, portable chest x-ray); propofol 40, fentanyl 150, RASS -3; WBC 18.4, procalcitonin 4.8, platelets 171, INR 1.4. Day shift I&O (EHR): 4,094 in, 474 out, +3,620 net.	Labs, I&O	0.5
Recommendation: titrate norepinephrine for MAP 65 or higher and call Dr. Brandt if the dose keeps rising (vasopressin above 0.25, hydrocortisone considered); lactate at 1900 and ABG; hourly glucose and urine output (call for under 20 mL/h); follow cultures and biopsy; mafenide soaks every 8 hours (next 0130) with pain premedication and acid-base watch; re-mark the erythema margin every 4 hours; hourly right-hand neurovascular checks; recheck K, Mg and phosphorus; central line and extravasation checks; family at the bedside.	Note 1850	0.5

PART 4: Shift 4: Tuesday 11/03 1900 to Wednesday 11/04 0700 (night shift) (16 pts)

Shift 4 is the sepsis peak and recovery: shock recognition, vasopressor math and weaning (graph), steroids and insulin, a critical culture call, and AKI with two nephrotoxic antibiotics. Watch for the traps: HIT is not the answer to the platelet fall, the 0200 output of 280 mL is mostly stool, and the pressor is never turned down in a hypothetical falling MAP. A careful student should earn about 12 of the 16 points.

4.1 The Peak of the Sepsis *[1 pts]*

Set the clock to 1930 on Tuesday. Copy the vital signs at each time, circle every abnormal value, then answer below: (a) Calculate the mean arterial pressure at 1930 from the blood pressure and compare it with the MAP on the monitor. (b) Why is a MAP of about 69 NOT reassuring in Marcus at 1930?

Vital	Tue 1930	Tue 2300	Wed 0300	Wed 0600
Temperature	102.4 °F	101.4 °F	100.5 °F	100.2 °F
Heart rate	133	117	108	101
Blood pressure	107/50	111/55	111/60	110/59
Respiratory rate	30	26	20	19
SpO2 (%)	92	96	98	97

Cells 0.75 x (cells correct / 20), rounded to the nearest 0.25; the two answers are 0.25 together (0.125 each). Allow +/- 2 for rates and +/- 0.2 F. Values: 1930: T 102.4, HR 133, BP 107/50, RR 30, SpO2 92; 2300: T 101.4, HR 117, BP 111/55, RR 26, SpO2 96; 0300: T 100.5, HR 108, BP 111/60, RR 20, SpO2 98; 0600: T 100.2, HR 101, BP 110/59, RR 19, SpO2 97. (a) MAP = (107 + 2 x 50) / 3 = 69 mmHg, which matches the monitor (69). (b) The MAP of 69 is only being kept above the goal of 65 by norepinephrine (0.10 at 1900, 0.12 at 1930, 0.14 from 2030), with fever 102.4, HR 133, RR 30 and lactate 4.1: he is in septic shock and the pressure would fall without the drug. The diastolic of 50 shows vasodilation.

4.1b MAP calculation and why 69 is not reassuring (scored with 4.1) *[0 pts]*

Model answer: Scored with item 4.1.

4.2 The 1845 Blood Gas *[0.5 pts]*

Open Lab Results (Blood Gas) and the ventilator settings for 1845. (a) Interpret the ABG in one sentence (acid-base disorder and what the body is doing about it) and name the lab drawn at 1900 that explains it. (b) Calculate the PaO2/FiO2 ratio and say what the chest x-ray at 1840 adds.

Required element (accept equivalent wording)	Where / who	Pts
pH 7.31, HCO3 18, base excess -8 = metabolic acidosis (PaCO2 36 is low-normal because he is breathing at 30 per minute: respiratory compensation, not a respiratory problem). Explained by the lactate of 4.1 at 1900 (lactic acidosis from shock); mafenide soaks can also add to a metabolic acidosis.	ABG 1845, lactate 1900	0.25
PaO2 92 / FiO2 0.50 = 184 (accept 180-190): a P/F under 200 on PEEP 8 is in the moderate ARDS range; the x-ray shows new bilateral perihilar/interstitial opacities, no consolidation = early ARDS physiology (capillary leak of sepsis), which is why the vent was changed to PEEP 8 and FiO2 0.50 at 1800 (set RR 18).	Vent, CXR	0.25

4.3 Norepinephrine: Rate and Safety *[1 pts]*

Open the MAR and the Orders. Norepinephrine 16 mcg/mL was 0.10 mcg/kg/min at 1900, 0.12 at 1930 and 0.14 at 2030. His dosing weight is 82 kg. (a) Calculate the pump rate in mL/h at 0.14 mcg/kg/min. (b) What MAP are you titrating to, and why is the arterial line and a central lumen used? (c) Give two safety checks for a vasopressor infusion.

Required element (accept equivalent wording)	Where / who	Pts
0.14 mcg/kg/min x 82 kg = 11.48 mcg/min x 60 = 688.8 mcg/h; / 16 mcg/mL = 43 mL/h (accept 43.0-43.1).	MAR	0.5
MAP 65 mmHg or higher (order: titrate by 0.02 mcg/kg/min every 5 minutes; the notes allow 0.02-0.04 every 5-10 minutes; accept either). A-line gives beat-to-beat pressure while the dose is changing; vasopressors are given through a central, dedicated lumen because they cause severe vasoconstriction and tissue injury if they leak into the skin.	Note 1755	0.25
Any two: independent double check of the concentration and rate; do not flush or bolus the line;		0.25

Score only what is written. Flag anything you cannot read for faculty review.

hourly site check of the left IJ (swelling, blanching) and stop/notify per protocol if infiltration is suspected; level and zero the transducer; never stop it abruptly; keep a back-up ready; titrate only per the order.		

4.4 Hydrocortisone, Glucose and the Insulin Infusion *[2 pts]*

Open the MAR, the Orders, Lab Results (point-of-care glucose and potassium) and the pharmacy note of 1930. (a) Hydrocortisone 50 mg IV q6h started at 2030: why would Dr. Brandt order a stress dose, and how many doses did you give this shift? (b) List the glucose values at 1930, 2130 and 0230 and the insulin rate changes between 1900 and 0300, and explain why the glucose rose after the steroid. (c) Why were the insulin rates lowered, and what is the hypoglycemia risk in Marcus? (d) Which electrolyte do you watch with insulin, and what were his values?

Required element (accept equivalent wording)	Where / who	Pts
Septic shock that still needs norepinephrine at 0.1 or higher (0.14 at 2030): critical illness can cause a relative adrenal insufficiency; hydrocortisone helps the vessels respond to norepinephrine and shortens time on the pressor. Two doses given by the night nurse: 2030 and 0230 (next 0830); the order is 50 mg q6h for 3 doses (third due 0830), then 25 mg q6h until the ICU team stops it; pharmacy plans a taper once off pressors, so never stop it abruptly.	MAR, pharmacy note	0.5
Glucose 246 (1930), 251 (2030), 266 (2130, peak), then 257, 241, 225, 212, 197 (0230). Insulin 4 units/h (running since 1800), raised to 5 at 2100, 3 at 0000, 1.5 at 0300. Hydrocortisone (a glucocorticoid) and sepsis stress hormones cause insulin resistance and more glucose release from the liver; the glucose was already rising with the sepsis (227 at 1730) and the peak came about one hour after the first dose.	Labs, MAR	0.5
Rates are lowered as the glucose falls toward the goal 140-180 mg/dL and as the sepsis improves (less insulin resistance). Risk of hypoglycemia: insulin requirement falls quickly while the feeding is constant at 70 mL/h; if the tube feeding is held or stops (OR, residual, tube problem) the glucose can drop fast; check hourly and after every rate change; per the order: glucose under 100 or tube feeding held = hold the infusion and call the provider; glucose under 70 = notify and give D50 per protocol, recheck.	Order	0.5
Potassium: insulin moves potassium into the cells (hypokalemia). K 3.8 (1700), 4.0 (2300), 4.1 (0500): acceptable, but keep 4.0 or higher, replace per protocol, and check with each BMP.	Labs	0.5

4.5 The 2100 Labs and the 2130 Gram Stain *[1 pts]*

Open the CBC at 2100 (compare with 1555 and with admission) and the wound Gram stain at 2130. (a) What do a WBC of 21.8 with 14% bands tell you? (b) Platelets fell from 171 to 142 (and were 288 on admission) while he receives heparin 5,000 units SQ: what percent fall from admission is that, and what do you ask or check before you decide to hold the heparin? (c) What does "Gram-positive cocci in clusters" most likely mean, and do the antibiotics already cover it? (d) What does contact precautions mean for you today?

Required element (accept equivalent wording)	Where / who	Pts
WBC 21.8 (up from 18.4) with 14% bands (immature neutrophils, a "left shift") = the bone marrow is being driven hard by a serious bacterial infection; consistent with burn wound sepsis (segs 68%).	CBC 2100	0.25
(288 - 142) / 288 = 51% fall since admission (already 128 at 0500 = 56%). Pharmacy asks to assess for HIT when platelets fall more than 50%: do not simply hold or continue; report the number and ask the provider; Dr. Brandt judged HIT unlikely (timing under 5 days of heparin, sepsis explains it). Watch for bleeding, trend platelets with the next CBC, and look at INR/fibrinogen for DIC (INR 1.3, fibrinogen 350 at 0500).	Platelets, pharmacy note	0.25
Gram-positive cocci in clusters = Staphylococcus, most likely S. aureus (suspected MSSA; MRSA not excluded until the culture finishes); a Gram stain is preliminary and does not give the antibiotic susceptibility. Vancomycin plus pip-tazo already cover it, so no change was ordered; wait for the final culture and sensitivities. The lab called the result at 2130 and the nurse called Dr. Brandt with read-back; Dr. Okafor to be told.	Gram stain 2130, note 2140	0.25
Contact precautions (burn unit): single room, gown and gloves for room entry and all wound care (plus mask for dressing changes), dedicated equipment (stethoscope, cuff), hand hygiene before and after, clean/dirty zones, sign on the door; accept equivalent items (the order says gown and	Orders	0.25

gloves for all wound care and room entry, dedicated equipment). S. aureus spreads by hands and equipment.		

4.6 Fever and Cooling *[0.75 pts]*

Open the temperature trend and the MAR (acetaminophen) from 1900 to 0600. (a) Write the temperature at 1930, 2200, 0000, 0300 and 0600 and the times acetaminophen 1 g IV was given. (b) Why record the temperature before you give acetaminophen, and why can you not give all the credit for the falling fever to acetaminophen? (c) Name two cooling measures you used or could use in a burn patient, and one thing you avoid.

Required element (accept equivalent wording)	Where / who	Pts
T 102.4 (1930), 101.6 (2200), 101.0 (0000), 100.5 (0300), 100.2 (0600). Acetaminophen 1,000 mg IV at 2200 and 0600 (scheduled q8h).	Vitals, MAR	0.25
Acetaminophen lowers the temperature and can mask the true fever curve; record the temperature before the dose so the team sees the real trend. The fever was already falling before the 2200 dose (102.4 at 1930 -> 101.9 at 2100) and kept falling as the antibiotics, steroids and source control worked, so the fall is partly acetaminophen and partly a real response. (AST 96/ALT 84 at 0500: keep to 3 g/day.)		0.25
Any two: remove extra linens, lower room temperature, cool cloths to axillae/groin away from burns, fan; core temperature monitoring (bladder probe every 30 minutes). Avoid: ice baths or cold water on the burn wounds (heat loss and hypothermia) and shivering, which raises oxygen use and heart rate.	Note 1950	0.25

4.7 Mafenide Soaks *[0.5 pts]*

The right arm and chest are treated with mafenide 5% soaks every 8 hours (1730, 0130). (a) Why is the pain different with this agent, and how do you score it in Marcus? (b) Mafenide can cause an acid-base problem: which one, and which lab results from this night would you check?

Required element (accept equivalent wording)	Where / who	Pts
Mafenide penetrates eschar but causes burning pain on application; premedicate with the ordered analgesic 15 minutes before the soak (order: hydromorphone 15 minutes before dressing changes; the MAR shows 0.5 mg at 1915 for wound care discomfort, and none is charted before the 0130 soak, so a good answer notes the 0130 soak should be premedicated) and score pain with the CPOT because he is intubated and cannot report pain (CPOT 1/8 hourly after 1930).	MAR, CPOT	0.25
It inhibits carbonic anhydrase and can cause a metabolic acidosis (hyperchloremic): check ABG pH/HCO3 and the BMP bicarbonate and chloride (HCO3 18 at 1845, 20 at 2230, 22 at 0400; CO2 22, 21, 22 and chloride 107, 106, 106 on the BMPs, so no worsening acidosis); report a falling bicarbonate or rising chloride.	Labs	0.25

4.8 Sedation and Delirium Risk *[0.5 pts]*

Open the Orders (RASS goal), the flowsheet and the MAR. (a) What is the RASS goal for Tuesday, what was his RASS (charted 2000, 0000 and 0400) on propofol 40 mcg/kg/min and fentanyl 150 mcg/h, and why did Dr. Brandt accept the deeper sedation tonight? (b) Name two delirium risk factors he has and two things you do about them.

Required element (accept equivalent wording)	Where / who	Pts
Goal -1 to -2 on Tuesday; his RASS was -3 at 2000 and 0000 (deeper than the goal) because he was triggering the ventilator (RR 30) in septic shock with early ARDS (PEEP 8, FiO2 0.50) and needed comfort and synchrony; no SAT tonight (Dr. Brandt). When he improved, propofol was lowered to 30 and fentanyl to 100 at 0400 and RASS became -2.	Order, vitals, note	0.25
Any two risks: sepsis, deep sedation/fentanyl, mechanical ventilation, steroids, sleep loss, restraint, alcohol use (withdrawal window 48-96 h is Wed-Thu); actions: CAM-ICU every shift (negative at 2000, 0000, 0400), lowest effective sedation and analgesia first, lights/noise down at night, cluster care, reorient and talk to him, family at the bedside, avoid benzodiazepines, remove restraints when not needed.	CAM-ICU	0.25

4.9 Septic Shock That Is Not Responding *[1 pts]*

Choose all that apply.

Correct: Notify Dr. Brandt now with SBAR...; Check the arterial line (level, zero, compare with a cuff pressure)...; Expect orders for a second vasopressor... measured boluses...; Look for other causes of a falling pressure...; Recheck the lactate and glucose, keep hourly urine output...

0.2 per correct option selected; subtract 0.2 for each incorrect option selected (not below 0). Wrong options: turning the pressor down (it is the treatment, never reduce or stop it without an order); a peripheral IV for the main vasopressor (central lumen is the standard; only short term with protocol); unlimited fluid (burn patients and the early ARDS lungs, PEEP 8, edema 2-3+ and weight gain make fluid creep dangerous); holding the steroid and antibiotics (both are part of the shock treatment and are given on time).

4.10 The Critical Result at 0240 *[2 pts]*

Open Lab Results (Microbiology) and the nursing note of 0300. At 0240 the microbiology lab calls you. (a) What do you do during the call (what do you say and write down)? (b) Who do you notify, and in what time frame? (c) Write the SBAR you give with the real values. (d) Which line is involved, and what do you assess at the bedside?

Required element (accept equivalent wording)	Where / who	Pts
Write the result down (or enter it) with the caller's name and the time, read it back (patient name and second identifier, blood culture set 2 from the central line, bottle 1 of 4 positive at 10 hours, Gram-positive cocci in clusters) and get the caller to verify it ("read back verified"); document the time of the call (0240).	Note 0300	0.5
Dr. Brandt (intensivist) was called at 0245; also tell the charge nurse and make sure Dr. Okafor (burn surgeon) and pharmacy (A. Chen) are aware; critical results go to a provider immediately (within the policy time, usually 15-30 minutes at most) and the response and any order are documented.	Note 0300	0.5
S: critical blood culture, 1 of 4 bottles, central line set, GPC in clusters. B: burn wound sepsis, cultures drawn 1550 before antibiotics, vancomycin and pip-tazo since the evening, wound Gram stain also GPC in clusters at 2130, left IJ line day 3. A: about 0200-0300 T 100.5-100.7, HR 107-108, BP 109-111/55-60 (MAP 73-77), lactate 1.8 (0200), norepinephrine 0.08, improving; line site clean. R: ask whether to continue vancomycin and pip-tazo (next vancomycin 0430), when to repeat blood cultures, and who decides about the line once the organism is identified (ID consult).		0.5
Left IJ triple lumen (placed Monday 0345): dressing dry, intact and dated; look and feel for erythema, drainage, tenderness and neck swelling; flush each lumen and check blood return; hub scrub 15 seconds; review whether it is still needed (norepinephrine and antibiotics): it stays until the team decides; do not remove it on your own. One positive bottle can be a contaminant, but with a matching wound Gram stain and S. aureus likely, treat it as real until the identification returns.	Note 0300	0.5

4.11 Reading the Pressor Graph *[2 pts]*

Study the graph of MAP, norepinephrine dose and lactate for this shift. (a) What happened to the MAP while the dose was raised from 0.10 to 0.14 and then lowered to zero? (b) How did the lactate change, and what does that mean? (c) What did you need to see before it was safe to wean, and who sets the criteria? (d) After the pressor is off, what do you watch, and what would make you call the provider?

Required element (accept equivalent wording)	Where / who	Pts
From 1900 to about 2130 the MAP stayed flat at 69-71 even though the dose was raised 0.10 -> 0.12 (1930) -> 0.14 (2030): more drug was needed just to hold above 65 (worst of the sepsis). By 2300-0000 the MAP climbed to 73-75 on the same 0.14 (antibiotics, steroid and albumin working). From 0000 the dose was stepped down (0.12, 0.08 at 0200, 0.04 at 0400, off 0430) and the MAP stayed 73-78, never below 65: no fall off the pressor.	Graph	0.5
Lactate 4.1 (1900) -> 2.6 (2230) -> 1.8 (0200) -> 1.5 (0500): a fall of about 37% in 3.5 hours ((4.1 - 2.6) / 4.1), normal (under 2) by 0200. A falling lactate means the tissues are getting enough blood flow and oxygen (less anaerobic metabolism); a rising lactate would mean worse shock.	Labs	0.5
MAP at or above goal (65, here 73-77) on a stable, low dose; lactate falling or normal; urine output at least 0.5 mL/kg/h (41 mL/h for 82 kg; he made 51-68 mL/h); warm skin, brisk capillary refill, heart rate coming down (133 to 108), no new fluid bolus, infection treated. The criteria come from the provider: the order says wean by 0.02 mcg/kg/min every 15-30 minutes when MAP is over 75, and Dr. Brandt's plan (2345) says once MAP is 70 or higher and lactate under 2. The first step down at 0000 was made with MAP 74-75 and lactate 2.6 (falling; 1.8 at 0200); accept any answer	Note 2345	0.5

Score only what is written. Flag anything you cannot read for faculty review.

that names a MAP above goal with a falling lactate. The nurse weans only per the provider's order, one small step at a time, and stops and calls if the MAP drops under 65.		
Recheck MAP every 15 minutes for the first hour, then hourly (done after 0430: MAP 77, 75, 76); heart rate, urine output, capillary refill/skin temperature, mental status, temperature and the next lactate; hydrocortisone continues. Call Dr. Brandt for MAP under 65, a rising heart rate, falling urine output, cool mottled skin or a rising lactate: the pressor may need to restart; keep the central line and the drug available.	Note 0450	0.5

4.12 The Kidneys, the Vancomycin Level and Procalcitonin *[1 pts]*

Open Lab Results (Creatinine, Vancomycin level, Procalcitonin) and the pharmacy note. (a) Write the creatinine at 1700, 2300 and 0500 and say what change this is (and whether his urine output agrees). (b) Why does the combination of vancomycin and piperacillin-tazobactam worry the pharmacist? (c) The vancomycin level of 11.6 was drawn at 0415: why is it drawn before the 0430 dose, and what question do you ask? (d) What does procalcitonin 9.6 add?

Required element (accept equivalent wording)	Where / who	Pts
Creatinine 1.1 (1700) -> 1.3 (2300) -> 1.4 (0500; baseline about 1.0): a rise of 0.3 mg/dL in 12 hours = acute kidney injury, stage 1 (KDIGO); BUN 27 -> 30. Urine output actually improved (0.34 mL/kg/h at 1900-2000 to 0.79 at 0500-0600), so it is a non-oliguric injury; likely sepsis/hypoperfusion plus nephrotoxic drugs.	Labs	0.25
Vancomycin plus pip-tazo is associated with more AKI than either drug alone; both depend on the kidneys, so a falling clearance makes levels rise; pharmacy will reassess vancomycin when the culture returns (if MSSA, a beta-lactam).	Pharmacy note	0.25
Drugs with a narrow safe range are checked by levels; the level is drawn at 0415, 15 minutes before the dose, so it shows the lowest concentration and clearance (11.6; pharmacy uses AUC dosing, not a trough alone). Draw it on time, and ask pharmacy/provider to review the 0430 dose of 1,250 mg in light of the rising creatinine (1.3 at 2300) before or after it runs; avoid NSAIDs, contrast and other nephrotoxins; strict urine output.	Pharmacy note	0.25
Procalcitonin 9.6 ng/mL (4.8 at 1555): high, a marker of severe bacterial infection and sepsis; it supports the diagnosis and the trend is used later to decide when to stop antibiotics; it does not identify the organism.	Labs	0.25

4.13 The Night Shift Intake and Output *[0.75 pts]*

Open the Intake/Output flowsheet (Totals) and the albumin order. (a) Write the totals for "Night shift 11/03 1900 to 11/04 0700" and the net balance, and (use the printed shift total, not your own sum of the hourly rows; ignore the later day-shift row) and compare the weight at 0400 (91.2 kg) with the admission weight (82 kg). (b) The output for 0200-0300 is 280 mL: how much is urine and what is the rest? (c) Albumin 5% 500 mL ran 2200 to 0200: what mL/h is that, why was it given, and what do you watch for?

Required element (accept equivalent wording)	Where / who	Pts
Intake 4,160 mL, output 803 mL, net +3,357 mL (4,160 - 803). Weight 91.2 kg vs 82 kg = +9.2 kg (about 11% above admission): fluid (edema 2-3+) from the capillary leak; watch lungs, ventilator pressures, edema, bladder pressure.	I&O totals, weight	0.25
Only 60 mL of the 280 mL at 0200-0300 is urine; the rest (220 mL) is a loose stool (tube-feeding related), so do not call it a diuresis. Real urine output improved from 28 mL/h (1900-2000, 0.34 mL/kg/h) to 68 mL/h (0600-0700, 0.83 mL/kg/h).	I&O hourly	0.25
500 mL / 4 h = 125 mL/h. Albumin 5% is a colloid: it draws and holds fluid in the vessels, supports the pressure and urine output in septic shock and burn leak with less crystalloid (LR was lowered to 75 mL/h at 0200). Watch for fluid overload: crackles, rising peak/plateau pressures, SpO2, edema, CVP, and for a BP rise.	MAR, order	0.25

4.14 Handoff at 0650 *[2 pts]*

It is 0650 on Wednesday. Write your SBAR handoff for J. Morales, RN, the day nurse. Use real numbers, the events of the night, what is due in the next hours (look at the MAR and Orders), and anything you want the day team to follow up.

Required element (accept equivalent wording)	Where / who	Pts
Situation: Marcus Delaney, BICU-6, full code, NKDA, contact precautions; burn wound septic	Note 0650	0.5

Score only what is written. Flag anything you cannot read for faculty review.

shock (MSSA suspected), now improving; off norepinephrine since 0430 (MAP 75-77); intubated.		
Background: 22% TBSA flame burn with inhalation injury, hospital day 3; cultures 1550 Tuesday before pip-tazo and vancomycin; hydrocortisone 50 mg q6h since 2030; wound Gram stain GPC in clusters (2130) and blood culture bottle positive GPC in clusters at 0240 (identification pending); left IJ line day 3, site clean.		0.5
Assessment: T 100.2 (from 102.4), HR 101, BP 110/59 (MAP 76), RR 19, SpO2 97%; vent AC/VC 450, RR 16, PEEP 5, FiO2 0.40 since 0600; lactate 4.1 -> 1.5; WBC 19.2 (bands 9%), platelets 128, creatinine 1.4 (peak), PCT 9.6, vancomycin level 11.6; UOP 65 mL/h (0.79 mL/kg/h); glucose 175 on insulin 1.5 units/h; propofol 30, fentanyl 100, RASS -2, CPOT 1; I&O +3,357; weight 91.2 kg.		0.5
Recommendation: MAP every hour and call for under 65; SAT/SBT at 0800; hydrocortisone and pip-tazo due 0830; mafenide soak due about 0930 with premedication; follow creatinine, urine output, platelets and glucose (hourly; hypoglycemia risk as insulin needs fall); pharmacy review of vancomycin; ID/burn team about the culture and the line; wean sedation/FiO2; acetaminophen next 1400 (LFTs); family updated, phone numbers on file.		0.5

PART 5: Shift 5: Wednesday 11/04 0700 to 1900 (day shift) (16 pts)

Shift 5 is the recovery day: weaning from the ventilator, the first hours after extubation, reading microbiology and narrowing antibiotics, moving from IV infusions to oral care, and a handoff. A careful student should earn about 12 of the 16 points. Notes on chart quirks: the microbiology results are filed in the Lab Results grid under the time each became available (MSSA at 1500, urine and repeat cultures at 1600); the I&O day-shift totals are those shown in the EHR (1,823 / 836 / +987).

5.1 The 0800 Sedation Break and the 0930 Breathing Trial *[1.5 pts]*

Read the 0820 nursing note, the 0955 respiratory therapy note and the SBT order. (a) What does a PASSED daily sedation interruption (SAT) look like, and what did Marcus show (RASS, CAM-ICU, pain, vital signs)? Which sedative was turned off and which continued? What would a FAILED SAT look like and what do you do? (b) Describe the spontaneous breathing trial (settings and length), then CALCULATE the RSBI from the respiratory rate and the exhaled tidal volume in the RT note and compare it with the pass value. (c) What did the cuff-leak test show and why is it done in a patient with an inhalation injury?

Required element (accept equivalent wording)	Where / who	Pts
Passed SAT: wakes to voice within minutes and stays calm and comfortable: RASS 0 (goal 0 to -1), follows commands (left hand, toes, nods), CPOT 0/8, CAM-ICU NEGATIVE, HR 90s, MAP 76-78, RR 17-18, SpO2 98%, no dyssynchrony. Propofol turned off at 0800; fentanyl 100 mcg/h continued for pain.	0820 note	0.5
Failed SAT: sustained agitation or anxiety, pain not controlled, RR over 35 or distress, SpO2 under 88-90%, new arrhythmia or ischemia, or a rising HR/BP. Action: stop the test, treat pain and distress, restart the sedative at HALF the previous rate (order), notify the provider, and try again the next day.	SAT order	0.25
SBT: pressure support 5 / PEEP 5 / FiO2 0.35 for 30 minutes (0925/0930 to 0955) with RT. RSBI = RR / VT in liters = 18 / 0.310 L = about 58 breaths/min/L (accept 56-60); pass is under 105. Other pass criteria met: RR under 30, SpO2 97%, HR change under 5% (limit 20%), no distress.	RT note, order	0.5
Cuff leak: audible leak, measured about 190 mL (more than 110 mL). A leak around the deflated cuff means the airway is not swollen shut; it is done because inhalation injury and 2 days of fluid resuscitation cause upper-airway edema, and without a leak there is a high risk of stridor and reintubation after extubation. (Still a concern in this patient even with a leak.)	RT note	0.25

5.2 Reading the Breathing Graph Before and After Extubation *[1.5 pts]*

The graph shows respiratory rate, SpO2 and heart rate from the flowsheet, Wednesday 0400 to 1200 (the dotted line is 30 breaths/min). He was extubated at 1045. (a) Looking only at 0600 to 1000, what does the graph show about his readiness to come off the ventilator? Use numbers. (b) What changes on the graph after 1045, give the values, and explain why this is or is not worrying. (c) Which graph changes would make you call the provider and RT, and what would you do at the bedside while waiting?

Required element (accept equivalent wording)	Where / who	Pts
Ready: from 0600 to 1000 the respiratory rate is steady and low (17-19), well under the 30 line; SpO2 is 97-99% on FiO2 0.40 then 0.35; heart rate is falling and settled (101 -> 93). No tachypnea, desaturation or tachycardia (the breathing trial was at 0925-0955, between the 0900 and 1000 points), so he is stable on minimal support (consistent with passing the SBT).	Graph, flowsheet	0.5
After 1045: RR rises from 18 to 20, 22 and 23 (plateau 23 at 1115-1200), SpO2 falls from 99% to 96-97%, HR 95 -> 91 (not rising). Expected and acceptable: he now breathes without ventilator pressure support or PEEP, has a sore, hoarse throat and is anxious, and gets 4 L nasal cannula (about 0.36 FiO2). It plateaued (RR still under 30, SpO2 at or above 92%, HR not rising), the ABG at 1200 was pH 7.39, PaCO2 41, PaO2 88, and there was no stridor or accessory muscle use at 1100 (the 1200 flowsheet notes only mild increased effort, hoarse voice, no stridor). Keep trending.	Graph, ABG 1200	0.5
Call for: RR continuing up toward or over 30 (or a steady climb), SpO2 under 92% despite more oxygen, a rising heart rate, stridor, accessory muscle use or tiring, falling mental status. At the bedside: stay with him, sit him upright (HOB 45 degrees or more), check airway sounds and work of breathing, increase O2 as ordered, call RT and Dr. Brandt, have suction, bag-valve mask and the reintubation kit ready.	Extubation note	0.5

5.3 Right After the Tube Comes Out (1045) *[1.25 pts]*

Open the 1100 Extubation Procedure Note and your 1130 nursing note. (a) List FOUR things you assess in the first hour after extubation (and how often) and what you found at 1050-1115. Why is he at higher risk for airway swelling than another patient? (b) If STRIDOR develops, what do you do and what does the provider plan say to give? (c) Why must the airway watch continue even though he looks well at 1100?

Required element (accept equivalent wording)	Where / who	Pts
Any four: stridor (high-pitched sound on breathing in), voice quality (hoarse but phonating is expected), work of breathing (accessory muscles, retractions, nasal flaring), RR, SpO2, HR/BP, cough strength and secretions, drooling or swallowing, mental status. Plan: vitals and airway assessment every 15 minutes for 1 hour, then hourly, continuous SpO2 and cardiac monitor. Findings: hoarse voice, weak but effective cough, no stridor, no drooling, no accessory muscle use, RR 20-23, SpO2 96-97% on 4 L.	Extubation note	0.5
Higher risk: grade 2 inhalation injury (swelling of the airway lining from smoke), about 56 hours of intubation (laryngeal trauma) and 2 days of large-volume resuscitation (edema).	Notes	0.25
Stridor: stay with him, sit him up, call RT and Dr. Brandt immediately (reintubation readiness), give oxygen, keep suction, bag-valve mask and reintubation equipment (8.0 and 7.0 tubes, video laryngoscope) at the bedside. Ordered treatment: racemic epinephrine 2.25% 0.5 mL in 3 mL saline by nebulizer, repeat every 20 minutes up to 3 times; heliox if available; hydrocortisone continues. Watch for a rising heart rate from the epinephrine and for swelling returning when it wears off. Keep NPO.	Order/plan	0.5

5.4 Dexmedetomidine After Extubation *[0.75 pts]*

Dexmedetomidine was started at 1100. Open the order. (a) Calculate the pump rate in mL/h (weight 82 kg, bag 4 mcg/mL, dose 0.4 mcg/kg/h). (b) When do you hold it and notify the provider according to the ORDER? (c) The surgeon wrote "avoid benzodiazepines": give two reasons that a drug like dexmedetomidine suits this patient better than lorazepam.

Required element (accept equivalent wording)	Where / who	Pts
0.4 mcg/kg/h x 82 kg = 32.8 mcg/h; 32.8 / 4 mcg/mL = 8.2 mL/h.	Order	0.25
Hold and notify for HR under 50 or SBP under 90 (as written in the order; the 1130 nursing note lists HR under 50 or MAP under 65: confirm against the order). Titrate 0.1 mcg/kg/h to RASS 0 to -1. Expected effects are bradycardia and hypotension: HR stayed 88-94 and MAP 78-81 for the rest of the shift.	Order	0.25
Any two: keeps him calm yet arousable and breathing on his own (little respiratory depression, important right after extubation); benzodiazepines raise the risk of delirium and prolong confusion in the ICU (CAM-ICU was negative and we want it to stay that way); benzodiazepines blunt his cough and breathing drive and add to opioids; Dr. Okafor and the ICU plan avoid them.	Notes	0.25

5.5 NPO, the OG Tube, Breathing Exercises and Nausea *[0.75 pts]*

After extubation he was NPO, the OG tube was removed at 1115, and at about 1215 he felt nauseated; ondansetron 4 mg IV was given at 1230. (a) Why NPO until the swallow screen, and why was the OG tube taken out now? (b) What do you do for incentive spirometry and oral care? (c) For the nausea: what do you do first, what is the safe-practice trap, and what do you check about the drug?

Required element (accept equivalent wording)	Where / who	Pts
NPO: after 56 hours of intubation the swallow and cough are weak, so food, water or saliva can go into the lungs (aspiration pneumonia); the screen is ordered before anything by mouth. OG tube out: feeds were stopped at 1030 and it is no longer needed once extubated; the tube sits across the throat and swallowing muscles, increases reflux and aspiration, irritates the nose and throat and is uncomfortable; he is awake. (He is NPO, so medicines are IV until the screen.)	Orders, notes	0.25
Incentive spirometer hourly while awake (slow deep breath, hold 3-5 seconds, about 10 times, then cough; coached to about 1,000 mL) to prevent atelectasis; HOB 45 degrees; oral care with swabs and a moist mouth because he cannot drink (reduces bacteria that could be aspirated).	1250 note	0.25
Nausea: sit him upright, check airway, suction ready, assess for cause (pain, swallowed secretions, anxiety), give ondansetron 4 mg IV PRN (order every 8 hours); no sips, ice chips or crackers for nausea because he is NPO until the screen (the trap); reassess in 30 minutes (improved at 1245).	MAR	0.25

Score only what is written. Flag anything you cannot read for faculty review.

Check QTc/other QT-prolonging drugs and last dose time before giving.		

5.6 The Insulin Infusion Ends at Noon: Reading the Correction Scale *[1 pts]*

Open the glucose results, the insulin lispro correction-scale order and the MAR. (a) His glucose values from 0930 to 1800 and why the infusion was stopped at 1200. (b) Using the ORDER, how many units of lispro would you give for glucose 149, 162, 238, 276 and 362, and when do you call the provider? (c) Name one safety concern in the hours around the change (feeds stopped, hydrocortisone dropped to 25 mg) and how often glucose is checked now.

Required element (accept equivalent wording)	Where / who	Pts
POC glucose 162 (0930), 154 (1030), 150 (1130), 149 (1800); serum 147 at 1700. The infusion (1 unit/h) was stopped at 1200 because glucose was stable around 150 (goal 140-180), the tube feeds had been off since 1030, the sepsis was resolving and the steroid dose was being reduced; he moves to subcutaneous correction insulin.	Labs, order	0.25
Scale: 151-200 = 2 units; 201-250 = 4; 251-300 = 6; 301-350 = 8; over 350 = 10 units AND call the provider. So 149 = 0 units (below the first row), 162 = 2 units, 238 = 4 units, 276 = 6 units, 362 = 10 units plus call. Under 70: treat per hypoglycemia protocol and call. Check the glucose first, pick the row that contains it, and read the number carefully (150 and 149 get no insulin).	Order	0.5
Concern: NPO or poor intake with insulin on board can cause hypoglycemia (feeds off since 1030, infusion still running until 1200; hydrocortisone being tapered lowers insulin need). Glucose is checked every 6 hours while NPO, then before meals and at bedtime once he eats (soft diet 1800); give correction insulin with the meal, not alone; watch for low-sugar signs.	Order	0.25

5.7 The 1330 Family Meeting *[0.5 pts]*

Read the Case Management family meeting note (the meeting was at 1330). (a) Give two things Dr. Okafor told the family about the next days. (b) The patient was not in the room: what happened afterwards, and give one thing you as the bedside nurse can do to support Marcus and his family after the meeting.

Required element (accept equivalent wording)	Where / who	Pts
Any two: continue burn ICU care 1-2 more days then burn step-down; excision and skin grafting tentatively Friday 11/06 (right forearm, hand and chest) with grafts from his own thigh; hospital stay about 2-4 weeks; prognosis good for survival, hand function depends on early therapy; rehab vs home with outpatient burn clinic decided after surgery; Full Code confirmed.	CM note	0.25
Dr. Okafor updated Marcus at the bedside afterwards and he agreed with the plan; consent for surgery is to be completed with the patient (not the family) on Thursday. Nurse support: reinforce the information in plain words, answer questions or relay them to the team, encourage the family to rest, support splint/exercise teaching, child life and social work (Ava), document the meeting and the patient's understanding.	CM note	0.25

5.8 The 1400 Dressing Change with Mafenide *[1 pts]*

Read the 1430 nursing note and the MAR. (a) When and what did you give before the dressing change, and why that timing? He still reported 6/10: what else do you do? (b) Why is mafenide used for this wound, and what two problems do you watch for with mafenide (find the numbers that show whether the second one is happening)? (c) Name four things you assess and document about the wound at this change, using what the note found (and why the marker line matters). (d) What precautions apply?

Required element (accept equivalent wording)	Where / who	Pts
Hydromorphone 0.5 mg IV at 1345, 15 minutes before the 1400 change so the IV peak effect (about 10-20 min) covers the pain; scheduled acetaminophen 1 g IV at 1400. Pain was 6/10 during, 4/10 after. Also: coach slow breathing, pause when asked (paused twice), warm/gentle technique, reassess 30 minutes after, sedation level (RASS) because of dexmedetomidine and opioid together.	MAR, note	0.25
Mafenide 5% soaks penetrate eschar, so they work on a deep wound infection (MSSA under the eschar) where silver sulfadiazine did not. Watch for (1) severe burning pain on application, and (2) metabolic acidosis because it inhibits carbonic anhydrase (also rash, hyperventilation). Numbers: bicarbonate 22 at 0500 and 24 at 1700, anion gap 9, ABG 1200 pH 7.39 / HCO3 24, RR not increasing, so no acidosis. Next soaks 1730; BMP every 12 h.	Order, labs	0.5
Any four: drainage amount and character (thin serosanguineous, no pus), odor (much less), erythema margin versus the skin-marker line from Tuesday 1430 (not extending: a spread means	Note	0.125

Score only what is written. Flag anything you cannot read for faculty review.

worsening infection), eschar edge (still separating at the right chest border), the violaceous area (unchanged, surgeon evaluating), face, right hand (Doppler signals, cap refill), photographs and measurements.		
Contact precautions with the burn nurse (N. Ibarra): gown and gloves for all wound care (per the order; add mask or eye protection if splash is likely, per unit policy), clean or sterile technique per unit policy; splint reapplied after.	Order	0.125

5.9 Reading the Culture Results in the Lab Section *[1.25 pts]*

The picture shows the Microbiology rows of Lab Results at 1630 Wednesday (cultures were drawn Tuesday at 1550). (a) For the wound culture, the blood cultures (4 bottles), the tracheal aspirate, the urine culture and the repeat blood cultures, write what each shows and the time column it appears under. Is any result still not final? (b) The rows for blood culture set 1, set 2 and the wound Gram stain look empty: why, and where would you scroll to find their results? (c) The lab called at 1500: what does "MSSA in 2 of 4 bottles" tell you (true infection or contaminant?) and what does it NOT yet tell you? (d) What does "no growth to date" on the repeat blood cultures mean at 1630?

Required element (accept equivalent wording)	Where / who	Pts
1500 column: wound culture = Staphylococcus aureus (MSSA) identified, susceptibilities to follow (NOT final); blood cultures (4 bottles) = MSSA in 2 of 4 bottles (one peripheral, one central line), preliminary; tracheal aspirate = normal respiratory flora, few WBC. 1600 column: urine culture = no growth at 24 hours (preliminary); repeat blood cultures = collected, no growth to date. Still pending: wound susceptibilities (and the final blood culture panel).	Screenshot, lab grid	0.5
The grid files each result under the time it became available. Set 1, set 2 and the Gram stain only have entries earlier: the Tuesday 1550 columns (collected, in process), Tuesday 2130 (wound Gram stain: many WBC, GPC in clusters) and Wednesday 0240 (set 2 central line bottle 1 of 4 positive at 10 hours, GPC in clusters, critical call). Scroll left to see them. The four-bottle result replaces them at 1500.	Lab grid	0.25
S. aureus in blood from two sets (one peripheral) is a true bacteremia, never treated as a contaminant; the wound is the likely source (clean line site). "Methicillin-sensitive" means beta-lactams work. It does not yet give the final susceptibility panel (which antibiotics; wound culture susceptibilities still pending) and does not show clearance or whether the line is infected.	ID/Pharmacy notes	0.25
"Collected, no growth to date" 30 minutes after collection only means nothing has grown YET (cultures take 24-48 hours to call; a 24-hour no-growth result would be expected Thursday afternoon); it is not proof of clearance.	Lab grid	0.25

5.10 MSSA: Narrowing the Antibiotics, Repeat Cultures and the Central Line *[1.5 pts]*

Read the 1145 ID consult, the 1540 pharmacy note, the Orders and the 1645 nursing note. (a) Vancomycin was stopped at 1530: why, and which antibiotic is better for MSSA? (b) Why does narrowing antibiotics matter, and why is pip-tazo still running? (c) What did ID ask for about repeat blood cultures: how many, from where, when (you drew them at 1600 before the 1630 pip-tazo), and for how long will he need antibiotics? (d) What is the decision about the left IJ central line and what would change it? What do you do for the line meanwhile?

Required element (accept equivalent wording)	Where / who	Pts
Vancomycin stopped 1530 because the organism is MSSA, not MRSA; vancomycin is inferior to a beta-lactam for MSSA, and vancomycin plus pip-tazo is nephrotoxic (creatinine peaked 1.4). Better drug: cefazolin 2 g IV q8h (ordered 1640 by Dr. Raman, first dose 0610 Thursday).	ID, pharmacy	0.375
Narrowing = stewardship: use the most effective, narrowest drug; reduces kidney injury and side effects, C. difficile, resistant organisms and gut flora damage, and the drug that kills MSSA best gives better outcomes. Pip-tazo continues (0830, 1630, 0030) as a bridge for possible Gram-negative or mixed burn-wound flora until wound and biopsy cultures are final; then it is stopped.	Pharmacy note	0.375
Two sets (one peripheral, one from the central line) at 1600 before the next antibiotic dose, repeated every 24-48 hours until negative. If uncomplicated MSSA bacteremia with a controlled source and clearance: at least 14 days of IV therapy counted from the first NEGATIVE culture (longer, 4-6 weeks, if complicated); excision and grafting Friday is source control.	ID note	0.375
Line stays for now (wound is the likely source, clean site, he needs central access, it is his only IV after 1800). Remove and replace at a new site if repeat cultures stay positive at 48-72 hours, the	ID note, order	0.375

site becomes infected, or he becomes unstable again; TTE if cultures stay positive. Meanwhile: daily necessity review, check the site every shift, chlorhexidine dressing, scrub the hub 15 seconds, aseptic technique, report fever or redness.		

5.11 First Time Out of Bed with PT and OT *[0.75 pts]*

Read the 1520 PT note, the 1535 OT note and the OOB order. (a) What do you watch and measure during the first dangle, stand and chair transfer, and when do you stop? (b) Give three safe-handling points for the lines and the burned right arm. (c) What are the splint instructions from OT?

Required element (accept equivalent wording)	Where / who	Pts
Monitor HR, BP, RR, SpO2 and symptoms before, during and after (dizziness, fatigue, pain, shortness of breath); dangle and chair findings: HR 98, SpO2 95%, RR 22, mild dizziness that resolved in 2 minutes; after: HR 91, SpO2 96%, BP 112/64. Stop and return him to bed for SpO2 under 92%, HR over 130, drop in BP, chest pain, marked dizziness or distress. Premedicate 30-45 minutes before.	PT note	0.25
Any three: 2-person assist with a gait belt; RN manages the lines (left IJ, left radial A-line, left AC PIV, Foley) so nothing pulls; log-roll avoiding the burned chest and arm; right arm supported on a pillow or trough above heart level; bed alarm, non-skid socks, fall precautions (weak legs, sedating drugs); oxygen on; not during the dressing change window.	PT/OT note	0.25
Resting hand splint in position of function (wrist 20-30 degrees extension, MCP 70-90 flexion, IP extended, thumb abducted); on at all times except dressing change and therapy; check skin under the straps every shift for pressure; reapply after the mafenide soaks; gentle finger movements hourly while awake; elevate the arm.	OT note	0.25

5.12 The Swallow Screen, Diet and the New Pain Pills *[0.75 pts]*

At 1600 he passed the bedside swallow screen and the soft diet started at 1800. (a) What did the screen include and what are your aspiration precautions? When do you stop the meal? (b) You give oxycodone 5 mg at 1645: why 5 mg, and what do you check first (look at the order and the other sedating drugs)? (c) Gabapentin 300 mg three times daily was also ordered: why, and what do you watch for?

Required element (accept equivalent wording)	Where / who	Pts
Screen: alert and upright, oral exam, ice chips, teaspoons then 3 oz (90 mL) water without cough, wet voice or desaturation (SpO2 95-96%), puree and soft solid. Soft diet with thin liquids and supplements. Precautions: upright 90 degrees for meals and 30 minutes after, small bites and slow sips, supervised first meals, no straws at first, oral care before and after. Stop and call for cough, wet voice, stridor, increased work of breathing or SpO2 under 92%.	SLP note	0.25
Order: 5 mg for pain 4-6, 10 mg for 7-10; his pain was 4/10, so 5 mg. Check RASS and RR first: hold for RASS -2 or lower or RR under 10 (high-alert opioid); it adds to dexmedetomidine and gabapentin (sedation, respiratory depression), so reassess sedation and breathing 30-60 minutes later; give for pain after the screen so he can swallow it with puree.	Order	0.25
Gabapentin: for burn neuropathic pain and itch (spares opioid); causes sedation, dizziness and fall risk, adds to the other sedatives, dose adjusted for kidney function (creatinine 1.2, CrCl about 100). Watch RASS, dizziness when he stands, fall precautions; pharmacy advises holding for RASS -2 or lower.	Order, pharmacy note	0.25

5.13 Wednesday Labs: Is the Infection Getting Better? *[0.75 pts]*

Open Lab Results and look at the creatinine, WBC (with bands) and procalcitonin from Tuesday afternoon through the 1700 draw on Wednesday. Write the values for each and what each trend tells you about his kidneys and the infection.

Required element (accept equivalent wording)	Where / who	Pts
Creatinine 1.1 (Tue 1700) -> 1.3 (2300) -> 1.4 (Wed 0500, peak; mild AKI, KDIGO stage 1 from shock plus vancomycin and pip-tazo) -> 1.2 (Wed 1700): improving as perfusion recovered and urine output is about 70 mL/h; BUN 30 -> 24. Keep avoiding nephrotoxins (NSAIDs, contrast).	Labs	0.25
WBC 18.4 (Tue 1555) -> 21.8 (2100, bands 14%) -> 19.2 (0500, bands 9%) -> 15.6 (Wed 1700, bands 4%): falling, fewer immature cells: the body is responding to treatment and source control.	Labs	0.25
Procalcitonin 4.8 (Tue 1555) -> 9.6 (Wed 0400, peak) -> 6.1 (Wed 1600): rose after the sepsis	Labs	0.25

Score only what is written. Flag anything you cannot read for faculty review.

started and is now coming down, which supports a responding infection (it lags); it does not replace the wound exam, temperature and cultures; keep trending daily.		

5.14 Intake and Output: Where Did the Fluid Go? *[0.75 pts]*

Open the Intake/Output flowsheet. The Day shift 11/04 0700 to 1900 total is intake 1,823 mL, output 836 mL, net +987 mL. (a) Calculate his average urine output in mL/h and mL/kg/h (82 kg) and compare it with the Parkland goal of 0.5 mL/kg/h. (b) Why is the urine output high now, and why is the net still positive although the fluids were cut (LR 75 -> 50 mL/h at 1200, stopped at 1800)? (c) What would you watch or ask the team about next?

Required element (accept equivalent wording)	Where / who	Pts
836 mL / 12 h = about 70 mL/h; 70 / 82 = about 0.85 mL/kg/h (accept 0.8-0.9), well above the 0.5 mL/kg/h (41 mL/h) goal; net 1,823 - 836 = +987.	I&O	0.25
Capillary leak is closing and the shocked kidneys have recovered: the fluid that went into the tissues (edema; the stay so far is about +17 liters) moves back into the vessels and is excreted (mobilization phase); less LR and normal kidney perfusion allow diuresis. Net is still positive because intake (IV carrier, LR, feeds until 1030, first oral intake at 1800-1900) still exceeds output, and insensible losses and the open burn wounds are not measured. The net is much smaller than the +3,357 on the night shift.	I&O, orders	0.25
Watch: urine hourly with the Foley (still needed), daily weight (91.2 kg by bed scale at 0400, up from 82 kg on admission; not rechecked on the day shift), lung sounds and oxygen need, edema, creatinine/BMP, potassium and magnesium (loss with diuresis), blood pressure; ask whether the Foley is still needed and whether any fluid is wanted once he eats and drinks. Call for a drop below 0.5 mL/kg/h (about 41 mL/h) or a rise in creatinine.	Orders	0.25

5.15 Handoff at 1850 *[2 pts]*

It is 1850 on Wednesday. Write your SBAR handoff for E. Haddad, RN, the night nurse. Use real numbers from the chart, include what happened today, what the night nurse must do or watch in the next hours (look at the MAR and orders), and the things still pending.

Required element (accept equivalent wording)	Where / who	Pts
Situation: Marcus Delaney, BICU-6, Full Code, NKDA, contact precautions; 22% TBSA burn with inhalation injury, MSSA sepsis from the wounds resolving; extubated 1045 and stable on 2 L nasal cannula: 1800 T 99.0, HR 90, BP 113/65 (MAP 81), RR 21, SpO2 96%, pain 4/10, RASS 0.	Flowsheets	0.5
Background: off norepinephrine since 0430, SAT and SBT passed (RSBI 58), no stridor; MSSA in 2 of 4 blood bottles and wound at 1500; vancomycin stopped 1530; pip-tazo continues, cefazolin 2 g IV q8h starts 0610 Thursday; repeat blood cultures drawn 1600; swallow screen passed 1600, soft diet at 1800; left AC PIV out and LR off at 1800, saline 20 mL/h via the left IJ (only access); insulin infusion stopped 1200 and lispro scale.	Notes, MAR	0.5
Assessment: day shift intake 1,823, output 836 (about 70 mL/h), net +987; labs 1700: WBC 15.6, bands 4%, creatinine 1.2 (from 1.4), K 4.2, glucose 147; PCT 6.1; glucose 149 at 1800 (no insulin needed); pain 4/10 on acetaminophen IV, oxycodone 5 mg at 1645; dexmedetomidine 0.4 mcg/kg/h with HR 88-91, no bradycardia; wound: drainage down, erythema within the marker; right hand Doppler present; dinner tray pending.	I&O, labs	0.5
Recommendation / to do: mafenide soaks 1730 done, next 0130; hydrocortisone 25 mg 2030; pip-tazo 0030; first oral doses incl. gabapentin; watch for stridor/hoarseness and aspiration with meals (upright, supervised); glucose before meals and bedtime with the scale; hourly IS; sedation and falls (assist x2); follow repeat blood cultures, wound culture final and the surgery plan (OR Friday: consent with Jenna present, NPO per surgery); keep dexmedetomidine hold parameters; monitor temperature and creatinine.	MAR, notes	0.5

PART 6: Shift 6: Wednesday 11/04 1900 to Thursday 11/05 0700 (night shift) (16 pts)

Shift 6 is the quiet night of recovery: oral pain control and the nutrition plan, a delirium and withdrawal watch after sedation is off, the change to targeted antibiotics, the removal of the arterial line and Foley, and the decision that he is ready to leave the ICU. The graph and the transfer SBAR bring the whole stay together. A careful student should earn about 12 of the 16 points.

6.1 Vital Signs Through the Last Night *[0.75 pts]*

Set the clock to 1900 on Wednesday 11/04. Copy the vital signs at each time and circle every abnormal value. Then write two sentences below: (1) what changed in his breathing support and pain across the night, and (2) one thing the vital signs cannot tell you about the infection.

Vital	Wed 1900	Wed 2300	Thu 0300	Thu 0500
Temperature	99.0 °F	98.9 °F	98.7 °F	98.8 °F
Heart rate	89	87	84	82
Blood pressure	110/63	113/64	114/66	112/64
Respiratory rate	20	19	19	18
SpO2 (%)	95	96	96	96
Pain (0-10)	4/10	3/10	3/10	3/10

Score 0.75 x (cells correct / 24), rounded to the nearest 0.25. The two sentences are worth 0.25 (item 6.1b): (1) oxygen was 2 L/min nasal cannula until it was weaned to room air at 0300 (SpO2 95-96% throughout); pain fell from 4/10 (1900) to 3/10 (2100 on) on oxycodone, gabapentin and acetaminophen; (2) vital signs cannot show what the wound looks like (purulent drainage, odor, spreading erythema), the white count and procalcitonin, or whether the blood cultures are clearing; a normal temperature can also be masked by scheduled acetaminophen. Chart values: T 99.0, 98.9, 98.7, 98.8; HR 89, 87, 84, 82; BP 110/63, 113/64, 114/66, 112/64; RR 20, 19, 19, 18; SpO2 95, 96, 96, 96; pain 4, 3, 3, 3. Everything is in the normal range except the pain score and the low-grade temperature of 99.0 (accept "no abnormal values" with pain circled). Allow +/- 2 for rates and +/- 0.2 F.

6.1b Two sentences for item 6.1 *[0.25 pts]*

(1) What changed in his breathing support and pain across the night? (2) One thing the vital signs cannot tell you about the infection.

Required element (accept equivalent wording)	Where / who	Pts
Both sentences correct: (1) 2 L nasal cannula to room air at 0300 with SpO2 96% and pain 4 to 3/10; (2) vital signs do not show the wound, WBC/procalcitonin or the cultures (acetaminophen can mask a fever).	Flowsheet	0.25

6.2 Soft Diet and the Correction Scale *[0.75 pts]*

Marcus eats a soft diet with shakes and the insulin infusion is off. (a) Read the dinner intake in the 2000 flowsheet and the glucose values at 1800, 0000 and 0600; using the lispro correction scale order, how many units were due for each? (b) The last hydrocortisone dose was at 0230: how will that change his glucose, and why is the risk of low glucose still small on this scale? (c) He eats only about half of each tray: name two actions for his nutrition before Friday's surgery.

Required element (accept equivalent wording)	Where / who	Pts
Dinner: ate 50% of the soft tray. Glucose 149 (1800), 141 (0000), 141 (0600); the scale gives 2 units only for 151-200 (4 units for 201-250), so 0 units each time (none were given on the MAR).	Flowsheet, labs, order	0.25
Stopping hydrocortisone (stress-dose steroids raise glucose) means glucose should drift DOWN over the next day; the risk of hypoglycemia is small because only a correction scale (no basal insulin) is ordered and glucose is 141; recheck before meals/bedtime and treat under 70 per protocol; ask the provider whether to hold the scale when NPO after midnight Thursday.	Orders, MAR	0.25
Any two: calorie count and record the percent of each meal; offer the high-protein shakes TID (ordered); help him sit upright to eat (aspiration precautions, 90 degrees); small frequent snacks; dietitian follow-up for higher calories/protein for wound healing and the donor site; vitamin C and zinc per MAR.	Orders, dietitian note	0.25

6.3 Pain and Sleep Now That He Is Awake *[1.25 pts]*

Open the 2230 and 0315 nursing notes, the pain flowsheet and the MAR. (a) Which pain scale do you use now, and why is the CPOT no longer the right tool? Give his scores at rest and with turning or the soaks. (b) List the doses of oxycodone, gabapentin and acetaminophen he received overnight with times. At 0130 mafenide soaks hurt (5/10) and the oxycodone was given at 0215: what would you do differently for the next soak at 0930? (c) At 0615 you are about to give the acetaminophen tablet: what must you check on the MAR and the order before giving it, and what is his daily limit? (d) Name two ways you protected his sleep and one safety check you make after giving oxycodone with gabapentin.

Required element (accept equivalent wording)	Where / who	Pts
Numeric 0-10 self-report scale, because he is awake (RASS 0), oriented and able to report; CPOT is for patients who cannot self-report (intubated or sedated). Scores: 4/10 at 1900, 3/10 at rest from 2100, 5/10 with turning and the soak (up to 6/10 with care).	Flowsheet, note	0.25
Oxycodone 5 mg at 2100, 0215 and 0615; gabapentin 300 mg at 2200 and 0620; acetaminophen 1 g IV at 2200, then 1,000 mg tablet at 0615. Next time give the oxycodone 30-45 minutes BEFORE the 0930 soak (Dr. Okafor's plan), not after the pain starts.	MAR, Okafor note	0.5
Check that no other acetaminophen is due or already given: the IV order ended at 0500 and the tablet order starts at 0600 (give only ONE form); last dose 2200 (more than 8 h ago); total in the last 24 h is 3 g (1400, 2200, 0615) and the limit on his order is 3 g/day because AST is 78, so the next dose is not before 1400.	Orders, MAR, labs	0.25
Any two sleep measures: cluster care and turns, lights dimmed, door closed, noise reduced, earplugs offered, pre-medicate before turns. After oxycodone plus gabapentin: check RASS (hold for RASS -2 or lower), respiratory rate and SpO2, and reassess pain 30-60 minutes later.	Note 2230, pharmacy	0.25

6.4 Delirium and Alcohol Withdrawal Watch *[1.25 pts]*

His drinking history and an ethanol level of 42 mg/dL at 0240 on Monday were charted. Sedation is off, so days 2 to 4 after the last drink are the highest-risk window. (a) Name the four features of the CAM-ICU and say what makes it positive; what are his CAM-ICU and RASS results on this shift? (b) List four signs of alcohol withdrawal or delirium tremens you would look for and which vital signs would change first. (c) Three non-drug actions that prevent delirium tonight. (d) If he became agitated and tremulous at 0400, what would you do first and what would you ask the provider (look at what he was told to avoid)?

Required element (accept equivalent wording)	Where / who	Pts
CAM-ICU: (1) acute onset or fluctuating mental status, (2) inattention, (3) altered level of consciousness (RASS not 0), (4) disorganized thinking. Positive = features 1 AND 2 plus either 3 or 4. His CAM-ICU is negative at 2000, 0000 and 0400 with RASS 0 (alert and calm), and negative at 1200 and 1600 Wednesday.	Flowsheet	0.5
Any four signs: tremor, sweating, tachycardia, hypertension, fever, nausea, anxiety/agitation, insomnia, hallucinations, disorientation, seizure; heart rate and blood pressure rise first (HR 82-89 and MAP 79-82 now).		0.25
Any three: sleep protection/clustering care, lights and day-night cycle, reorientation, glasses/family contact and phone call, early mobilization, pain control, avoid benzodiazepines and extra sedatives, treat constipation, avoid restraints; thiamine and folic acid daily.		0.25
Stay with him and check safety, vital signs, glucose, pain, bladder (Foley out), oxygen, then CAM-ICU/RASS and CIWA-type withdrawal scoring (he can talk now); notify the provider (SBAR). Question: benzodiazepines (lorazepam) are on the list to avoid because of delirium; ask what the withdrawal order would be, and about thiamine before glucose.	Pharmacy note	0.25

6.5 Pip-Tazo to Cefazolin *[1.25 pts]*

The last pip-tazo dose is given at 0030 and cefazolin 2 g IV is given at 0610. (a) Compare the two drugs for this patient: what did the blood and wound cultures grow, which drug is the best choice for it, and give TWO reasons for de-escalating from pip-tazo (look at the pharmacy and ID notes and the allergy list). (b) Pip-tazo is an extended infusion: how long does the 0030 dose run? How long is the cefazolin infusion on the order, and what rate in mL/h gives 2 g in 100 mL over that time? (c) Write the next two cefazolin times, and name two things you check before and after the first dose.

Required element (accept equivalent wording)	Where / who	Pts
Blood cultures (2 of 4 bottles) and wound culture: MSSA; wound final heavy MSSA susceptible to cefazolin, nafcillin, clindamycin, TMP-SMX (resistant to penicillin); tracheal aspirate normal	Labs, notes	0.5

Score only what is written. Flag anything you cannot read for faculty review.

flora, urine no growth. Cefazolin, a first-generation cephalosporin, is a first-line, targeted drug for MSSA bacteremia and is better than vancomycin for MSSA.		
Any two reasons for de-escalation: pip-tazo is broad-spectrum and no Gram-negative or anaerobic organism was found, so narrowing lowers the risk of resistance and C. difficile; narrower, targeted therapy is as good or better for MSSA; less nephrotoxicity (creatinine peaked at 1.4 on vancomycin plus pip-tazo); fewer interactions; stewardship/ID recommendation. Allergies: none known, so a cephalosporin is safe.	Pharmacy note	0.25
Pip-tazo runs over 4 hours (0030 to about 0430). Cefazolin order: infuse over 30 minutes; 100 mL / 0.5 h = 200 mL/h.	Orders	0.25
Next doses 1410 and 2210 (q8h from 0610). Check the allergy list and ask about prior reactions (none known); watch for rash, itching, hives, wheeze or low BP during and after the infusion, for diarrhea (C. difficile) later, and the creatinine; flush the line before and after and do not mix with other drugs.	Pharmacy reconciliation	0.25

6.6 Three Treatments That Stopped Overnight *[1.25 pts]*

Three supports were stopped or weaned during the night. For each one, write when and how it stopped, what the chart showed afterward (look at the 0300 and 0500 vital signs and the notes), and what you watch for next.

Model answer: Rows worth 0.5 (hydrocortisone), 0.5 (dexmedetomidine), 0.25 (oxygen).

Required element (accept equivalent wording)	Where / who	Pts
Hydrocortisone: 50 mg q6h from Tue 2030 (3 doses: 2030, 0230, 0830), then 25 mg q6h from Wed 1430 (1430, 2030, 0230); the 0230 dose was the last and the order ended at 0600 (no further dose). Afterward BP 114/66 (MAP 82) and 112/64 at 0500, HR 82. Watch for low blood pressure, weakness, fever returning, and glucose falling (less insulin resistance).	MAR, orders, notes	0.5
Dexmedetomidine 0.4 mcg/kg/h since Wed 1100, off at 0300. Afterward RASS 0, calm, HR 84 then 82, BP 114/66, no rebound tachycardia or hypertension, observed for 4 hours. Watch for rebound (tachycardia, hypertension, agitation), withdrawal signs and sleep, since it also masks withdrawal signs.	MAR, note 0315	0.5
Oxygen: 2 L/min nasal cannula weaned to room air at 0300; SpO2 96% at 0300 and 0500, RR 18-19, lungs clear; chest x-ray at 0520 clear. Check SpO2 at rest and with turning every hour; resume oxygen for SpO2 under 92%; incentive spirometry.	Flowsheet, orders	0.25

6.7 The Three-Day Trend Graph *[2 pts]*

Study the graph of temperature, WBC, procalcitonin and creatinine for the whole stay (from the chart). Use the 0430 values on Thursday (WBC 11.2, creatinine 1.0, procalcitonin 2.1). (a) For each of the four lines, state the peak value and when it happened, and the latest value. (b) What does the graph show about the infection and about his kidneys? (c) At what time did he turn the corner, and why do the lab lines peak AFTER the temperature does? (d) What must the Med-Surg team keep watching? Give three things from the graph and the chart.

Required element (accept equivalent wording)	Where / who	Pts
Temperature peak 102.4 F at Tue 1930 (then falls to 100.4 by Wed 0400 and below it from 0500, 98.8 at Thu 0500; the last point on the graph is 98.5 at Thu 0700, accept 98.5-98.8). WBC peak 21.8 at Tue 2100 (now 11.2). Procalcitonin 4.8 (Tue 1600), peak 9.6 at Wed 0400, 6.1, now 2.1 at Thu 0430. Creatinine 1.1 baseline, 1.3 Tue 2300, peak 1.4 Wed 0500, 1.2, now 1.0 (admission 0.9). (Accept +/- a few hours.)	Graph, labs	0.5
Infection: a burn wound sepsis (MSSA bacteremia) shown by the fever, leukocytosis and rising procalcitonin that are now clearly resolving (WBC and PCT falling). Kidneys: creatinine rose 1.1 to 1.4 (a mild acute kidney injury from septic shock, vancomycin plus pip-tazo, and low BP) and has recovered to 1.0, near baseline; urine output stayed adequate.	Graph	0.5
Turning point: the temperature peaked about Tuesday 1930 and the WBC at 2100, about 3 to 4.5 hours after antibiotics (pip-tazo 1630, vancomycin 1645), fluids and norepinephrine (1730); lactate fell 4.1 to 2.6 (2230); norepinephrine was off at Wed 0430. Procalcitonin and creatinine lag behind: PCT rises for 12-24 hours and falls slowly, and the creatinine reflects kidney damage that heals over a day or two; so they peak on Wednesday morning even though the patient was already improving.	Graph, MAR	0.5

Score only what is written. Flag anything you cannot read for faculty review.

Any three: a fever of 101.5 F or higher, HR over 120 or MAP under 65 (call the provider); WBC and procalcitonin must keep falling (PCT 2.1 is still above normal); creatinine/urine output after the Foley is out and with cefazolin and mafenide (BMP daily, bicarbonate); repeat blood cultures (final at 5 days); the wound for purulence, odor, spreading erythema beyond the marked line; the central line site; hydrocortisone stopped (BP).	Graph, Okafor note	0.5

6.8 The Arterial Line and the Foley Come Out at 0500 *[1.25 pts]*

Read the 0520 nursing note. (a) Describe how you remove the left radial arterial line, how long you hold pressure (and why that could be longer in some patients), and what you check in the left hand and the site afterward and when. Which arm is used for the blood pressure now? (b) The Foley comes out at 0500: what is the target time for the first void, what do you do if he does not void, and what is his urine output goal in mL/h (0.5 mL/kg/h at 82 kg)? (c) Give two CAUTI prevention measures for the catheter days and one reason this one was removed now.

Required element (accept equivalent wording)	Where / who	Pts
Arterial line: wash hands and wear gloves, explain, stop the transducer/pressure bag, remove the dressing and securement, withdraw the catheter and immediately apply firm direct pressure for 5 minutes (longer if on anticoagulants or low platelets/high INR; his platelets are 186, INR 1.1, on SQ heparin), check that the catheter is intact, apply an occlusive dressing.	Note 0520	0.5
After removal: check the left hand for color, warmth, capillary refill (under 2 seconds), radial pulse, sensation and movement, and the site for bleeding or hematoma at 15 and 60 minutes (then per routine); BP is now by cuff on the LEFT arm (the right arm is burned and splinted).	Note 0520	0.25
Void within 6 hours of removal (by 1100); if no void by then or if he has suprapubic fullness/discomfort, do a bladder scan and call the provider for straight cath/orders; encourage fluids, urinal at bedside, record every void; goal 0.5 mL/kg/h x 82 kg = 41 mL/h (accept 30-41 mL/h with the reasoning; the burn surgery note asks for a call if UOP is under 30 mL/h).	Note 0520, orders	0.25
Any two CAUTI measures: closed drainage system, bag below the bladder and off the floor, securement device, catheter care each shift, daily necessity review, hand hygiene. Removed now because the resuscitation is over and hourly urine output is no longer needed (day 4 of the catheter).	Order	0.25

6.9 The Central Line on Line Day 4 *[0.75 pts]*

The left IJ central line stays after the arterial line, the PIV and the Foley are gone. (a) Count the line day (look at the Devices page) and explain why ID allowed it to stay even though he had bacteremia. (b) Give three CLABSI-prevention actions you do on this line every shift. (c) What is the one question you would ask the provider to get the line out sooner?

Required element (accept equivalent wording)	Where / who	Pts
Placed Mon 11/02 0345: Mon = day 1, so Thursday is line day 4. It stays because the repeat blood cultures (Wed 1600) show no growth so far (no growth to date at about 12 hours, preliminary); ID says remove it only if they turn positive; it is also his only IV access now.	Devices, notes	0.25
Any three: assess the site and dressing each shift (clean, dry, intact, dated chlorhexidine dressing); scrub the hub for 15 seconds before every access; minimize entries; change tubing per policy; dressing change every 7 days or if soiled; daily necessity review; hand hygiene; report fever, chills or site drainage and culture per order.	Order	0.25
Ask: now that he is off vasopressors and eating, can we place a peripheral IV and remove the central line, or what result tells us it is safe to remove it (repeat cultures final at 5 days)? A new PIV is needed on arrival.	Transfer summary	0.25

6.10 Is He Ready to Transfer? *[1.25 pts]*

The transfer to Med-Surg 2 East is planned for 0700. For each criterion, write the chart value or time that proves it is met (with the actual number of hours where asked), and in the last row state your decision (ready or not ready) and one concern you would still hand off.

Model answer: Each of rows 1-7 is 0.125 (criterion correctly evidenced), row 8 (decision and concern) 0.375.

Required element (accept equivalent wording)	Where / who	Pts
Off pressors more than 24 h: norepinephrine off Wed 0430, so 26.5 hours at 0700 (about 25.5 h at 0600); nothing vasoactive running.	MAR	0.125

Score only what is written. Flag anything you cannot read for faculty review.

Extubated more than 20 h: ETT out Wed 1045, so 20 h 15 min at 0700 (19 h at 0600; the 20-hour mark is 0645); no stridor, voice hoarse but improving.	Devices	0.125
Urine output adequate: 52-63 mL/h each hour from 1900 to 0500 (0.63-0.77 mL/kg/h) against the 41 mL/h goal.	Flowsheet	0.125
Hemodynamics stable: HR 82-89, BP 110-114/63-66, MAP 79-82 (1900 to 0500), no pressors, lactate 1.5 (Wed 0500), afebrile 98.7-99.0 F.	Vitals	0.125
Pain controlled on oral meds: 3/10 at rest on oral acetaminophen, gabapentin and oxycodone; no IV hydromorphone since Wed 1345.	MAR	0.125
Room air: since 0300, SpO2 96% (0300 and 0500), RR 18-19; chest x-ray (0520) clear.	Flowsheet	0.125
Burn team agrees: Dr. Okafor (0600 note) and Dr. Brandt approved the transfer; case management note 0545 confirms the bed.	Notes	0.125
Decision: ready (all seven criteria met, with strong trends). One concern: no peripheral IV (central line only), first void after Foley removal is pending (by 1100), withdrawal/delirium watch after dexmedetomidine, glucose off steroids, OR Friday. A student who says "not ready" must name a criterion that is not met.	Chart	0.375

6.11 Fall Risk and Mobility After the Lines Are Out *[0.75 pts]*

The arterial line and Foley are out, he is on gabapentin and oxycodone and his right arm is splinted. Which actions are correct for the rest of this shift and the handoff?

Correct: Keep the bed low, the bed alarm on...; Assist x2 with a gait belt...; Offer the urinal and a toileting schedule...; Teach him to call before getting up, and have him dangle...; Hand off the fall risk and the PT/OT plan...

0.15 per correct option selected; subtract 0.15 for each wrong option (not below 0). Wrong options: walking alone (fall precautions need assist x2); holding all pain/neuropathic medicine (pain control is a transfer criterion; give and check RASS instead); alarm off (alarm and precautions continue); restraint (no order, extubated, calm and oriented, not indicated).

6.12 Night I&O Totals *[0.75 pts]*

Open Intake/Output and read the line "Night shift 11/04 1900 to 11/05 0700". (a) Write the intake, output and net. Compute his average urine output in mL/h and in mL/kg/h (82 kg) over the 12 hours. (b) Compare this night with the night before (11/03 1900 to 11/04 0700) and say what the change means for his fluid status. (c) His weight is 89.8 kg this morning and 82 kg on admission: what does that tell you, and how will you track output now that the Foley is out?

Required element (accept equivalent wording)	Where / who	Pts
Night shift: intake 1,257 mL, output 638 mL, net +619 mL (as the flowsheet shows). 638 / 12 = about 53 mL/h; 53 / 82 = about 0.65 mL/kg/h (accept 0.6-0.7), above the 0.5 mL/kg/h (41 mL/h) goal. Accept 53-58 mL/h (0.65-0.71 mL/kg/h) if the student divides by the 11 hourly rows or uses the hourly values.	I&O totals	0.25
Previous night: 4,160 in, 803 out, +3,357. The net fell from +3,357 to +619 because the resuscitation fluids and albumin are finished (only the saline carrier and oral intake now); the capillary leak is over and the body is now holding less fluid; he is turning from fluid loading to fluid shifting back (edema 1-2+ and decreasing).	I&O totals	0.25
Weight is up 7.8 kg from admission: the extra water is in the tissues (edema), not new tissue, and should come off as the leak closes; track by the daily weight and edema. Without the Foley: measure each void in a urinal/hat, record the time and volume, bladder scan if no void by 6 hours, and keep intake and output charting.	Flowsheet	0.25

6.13 The Final Handoff: Transfer SBAR to Med-Surg *[2.5 pts]*

It is 0650 on Thursday 11/05. Write your SBAR transfer handoff to C. Pruitt, RN (Med-Surg 2 East, burn step-down). Use real numbers and include: the burn infection signs to watch, central line day, the cefazolin schedule, what is still pending, the pain plan, mobility, Friday's OR and the NPO time, and the family contacts. Mention one safety item for the first hours after arrival.

Required element (accept equivalent wording)	Where / who	Pts
Situation: Marcus Delaney, 34 M, MRN SIM-40101, Full Code, NKDA, contact precautions, transferring from BICU-6 to 2 East at 0700; stable: T 98.8, HR 82, BP 112/64, RR 18, SpO2 96% on room air, pain 3/10, RASS 0, CAM-ICU negative.	Notes, flowsheet	0.5
Background: house fire 11/02, 22% TBSA (right arm, chest, face/neck; about 8% full-thickness)	Notes	0.5

Score only what is written. Flag anything you cannot read for faculty review.

with grade 2 inhalation injury; intubated 0250 Mon to 1045 Wed; right forearm escharotomy; Parkland done; Tuesday MSSA burn wound sepsis with bacteremia, norepinephrine 1730 Tue to 0430 Wed, hydrocortisone stopped (last dose 0230); vancomycin stopped Wed 1530; pip-tazo last dose 0030.		
Assessment: WBC 11.2, creatinine 1.0, procalcitonin 2.1, platelets 186, K 4.3 (0430); glucose 141; urine output 52-65 mL/h, night I&O +619; wounds: mafenide soaks q8h (0930, 1730, 0130), erythema inside the marked line, no purulence or odor; right hand Doppler present; left IJ central line day 4 (placed Mon 0345), site clean, A-line and Foley removed 0500 (first void by 1100), no PIV so a new one is needed; eating about half of soft diet; weight 89.8 kg.	Labs, flowsheet	0.5
Recommendations: watch for burn wound infection (purulent drainage, foul odor, spreading erythema past the marker, new dark/violaceous areas or eschar change, more pain) and sepsis (T 101.5 F or higher, HR over 120, MAP under 65, UOP under 30 mL/h) and call Dr. Okafor; cefazolin 2 g IV q8h next at 1410 and 2210 (over 30 min); pending: final blood culture susceptibilities, repeat blood cultures from Wed 1600 (final at 5 days; no growth to date at about 12 h), wound biopsy histology and quantitative culture (the wound culture final resulted at 0600: MSSA, cefazolin-susceptible); pain: acetaminophen 1 g q8h (max 3 g/day), gabapentin 300 mg q8h, oxycodone 5-10 mg q4h PRN 30-45 min before soaks, hydromorphone 0.5 mg IV only for severe pain; mobility: assist x2, chair with PT/OT, right arm elevated and splinted, fall precautions and bed alarm; OR Friday 11/06 for excision and autografting: NPO after midnight Thursday night, consent with the patient and Jenna Whitaker, repeat type and screen this morning; family: Jenna Whitaker (partner, health care proxy, asked to be called at any hour), Linda Delaney (mother), daughter Ava age 6, K. Abara LCSW and child life.	Notes, orders	1

SCORE SHEET

Student: _____ Date graded: _____ Passing score: 78 of 100

Item	Description	Max	Earned
Part 1	**Shift 1: Monday 11/02, 0700 to 1900 (day shift)**	**20**	
1.1	Why Is He Here? Your Picture of Marcus at 0700	2.5	
1.2	The Parkland Formula	2	
1.3	Vital Sign Trend	1.5	
1.3b	Trend (scored with 1.3)	0	
1.4	The Pulse Oximeter Says 100%	1.5	
1.5	Titrating the Fluids	1.5	
1.6	Calling the Burn Surgeon	1.5	
1.7	Lines, Tubes and Airway	1.5	
1.8	The Ventilator and Sedation	1.5	
1.9	Reading the Labs	1.5	
1.10	Adding Up Your 12 Hours	1.5	
1.11	The 1400 Dressing Change	1.5	
1.12	Handoff at 1900	2	
Part 2	**Shift 2: Monday 11/02 1900 to Tuesday 11/03 0700 (night shift)**	**16**	
2.1	Vital Sign Trend Through the Night	1.5	
2.1b	Overall night, and what the vitals do not show (scored with 2.1)	0	
2.2	Glucose and the Insulin Infusion	2	
2.3	The End of the First 24 Hours	1.5	
2.4	Alcohol and the Next Three Days	1	
2.5	The Agitation Event at 0245	2	
2.6	Checking the Right Hand and the Abdomen	1.5	
2.7	Tube Feeding and Calories	1.5	
2.8	The Midnight and 0600 Labs	1.5	
2.9	Propofol: Math and Safety	1.5	
2.10	Handoff at 0650	2	
Part 3	**Shift 3: Tuesday 11/03 0700 to 1900 (day shift)**	**16**	
3.1	The 0800 Sedation Interruption	1	
3.2	Albumin and the Feeds	1	
3.3	Noon Electrolytes and the Replacement Orders	1.5	
3.4	The 1400 Dressing Change	1	
3.5	Analyze the Graph: Temperature, Heart Rate and MAP	1.5	
3.6	The SBAR Call at 1540	1.5	
3.7	The Sepsis Bundle: Cultures and Lactate	1.5	
3.8	Checking the Antibiotic Orders	1.5	
3.9	Low MAP and Norepinephrine	1.5	
3.10	Switching the Topical Antimicrobial	1	
3.11	Falling Urine Output	1	
3.12	Handoff at 1850	2	
Part 4	**Shift 4: Tuesday 11/03 1900 to Wednesday 11/04 0700 (night shift)**	**16**	

Score only what is written. Flag anything you cannot read for faculty review.

4.1	The Peak of the Sepsis	1	
4.1b	MAP calculation and why 69 is not reassuring (scored with 4.1)	0	
4.2	The 1845 Blood Gas	0.5	
4.3	Norepinephrine: Rate and Safety	1	
4.4	Hydrocortisone, Glucose and the Insulin Infusion	2	
4.5	The 2100 Labs and the 2130 Gram Stain	1	
4.6	Fever and Cooling	0.75	
4.7	Mafenide Soaks	0.5	
4.8	Sedation and Delirium Risk	0.5	
4.9	Septic Shock That Is Not Responding	1	
4.10	The Critical Result at 0240	2	
4.11	Reading the Pressor Graph	2	
4.12	The Kidneys, the Vancomycin Level and Procalcitonin	1	
4.13	The Night Shift Intake and Output	0.75	
4.14	Handoff at 0650	2	
Part 5	**Shift 5: Wednesday 11/04 0700 to 1900 (day shift)**	**16**	
5.1	The 0800 Sedation Break and the 0930 Breathing Trial	1.5	
5.2	Reading the Breathing Graph Before and After Extubation	1.5	
5.3	Right After the Tube Comes Out (1045)	1.25	
5.4	Dexmedetomidine After Extubation	0.75	
5.5	NPO, the OG Tube, Breathing Exercises and Nausea	0.75	
5.6	The Insulin Infusion Ends at Noon: Reading the Correction Scale	1	
5.7	The 1330 Family Meeting	0.5	
5.8	The 1400 Dressing Change with Mafenide	1	
5.9	Reading the Culture Results in the Lab Section	1.25	
5.10	MSSA: Narrowing the Antibiotics, Repeat Cultures and the Central Line	1.5	
5.11	First Time Out of Bed with PT and OT	0.75	
5.12	The Swallow Screen, Diet and the New Pain Pills	0.75	
5.13	Wednesday Labs: Is the Infection Getting Better?	0.75	
5.14	Intake and Output: Where Did the Fluid Go?	0.75	
5.15	Handoff at 1850	2	
Part 6	**Shift 6: Wednesday 11/04 1900 to Thursday 11/05 0700 (night shift)**	**16**	
6.1	Vital Signs Through the Last Night	0.75	
6.1b	Two sentences for item 6.1	0.25	
6.2	Soft Diet and the Correction Scale	0.75	
6.3	Pain and Sleep Now That He Is Awake	1.25	
6.4	Delirium and Alcohol Withdrawal Watch	1.25	
6.5	Pip-Tazo to Cefazolin	1.25	
6.6	Three Treatments That Stopped Overnight	1.25	
6.7	The Three-Day Trend Graph	2	
6.8	The Arterial Line and the Foley Come Out at 0500	1.25	
6.9	The Central Line on Line Day 4	0.75	
6.10	Is He Ready to Transfer?	1.25	

Score only what is written. Flag anything you cannot read for faculty review.

6.11	Fall Risk and Mobility After the Lines Are Out	0.75	
6.12	Night I&O Totals	0.75	
6.13	The Final Handoff: Transfer SBAR to Med-Surg	2.5	
	TOTAL	**100**	

Result: [ ] PASS (78 or more) [ ] DOES NOT PASS (below 78) Items needing faculty review (illegible or unclear): _____